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Meyer et al.

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(54) **WIRELESS-ENABLED SURGICAL SUTURE NEEDLE**

(65) **Prior Publication Data**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 773 days.

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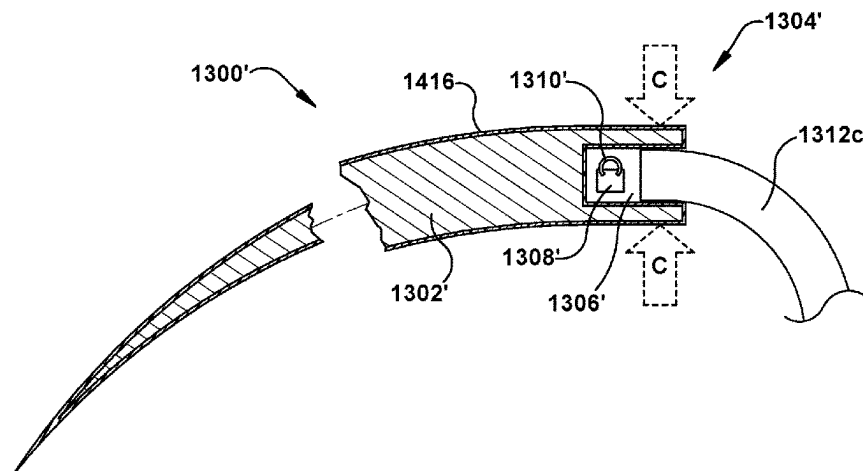
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(57) **ABSTRACT**

A suture needle apparatus includes a needle body and a radio-frequency identification ("RFID") chip disposed

(Continued)



proximate to the needle body. The RFID chip is encoded with an identifying information associated with the suture needle apparatus. The RFID chip includes an electromagnetic coupling element. A suture thread is operatively coupled to the needle body. At least one of the needle body and the suture thread is an antenna selectively electromagnetically coupled to the RFID chip and, when coupled, is configured to wirelessly communicate the identifying information responsive to radio-frequency interrogation of the suture needle apparatus. A method of providing a suture needle apparatus is also disclosed.

19 Claims, 12 Drawing Sheets

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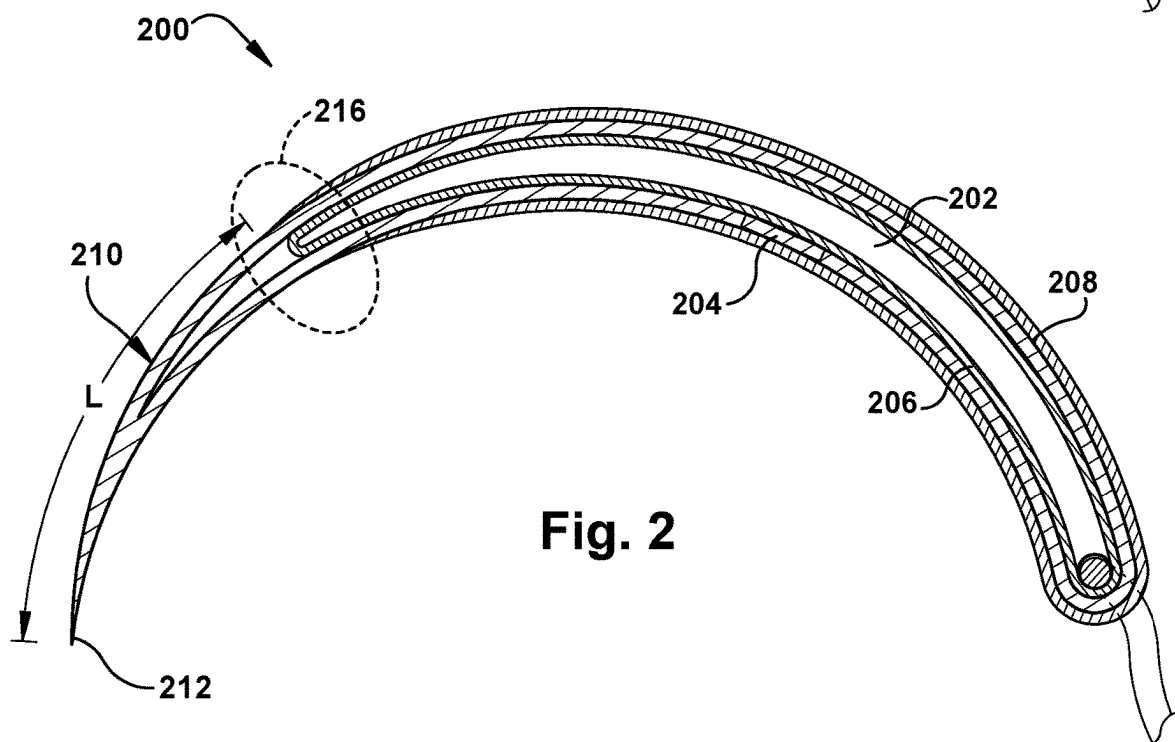
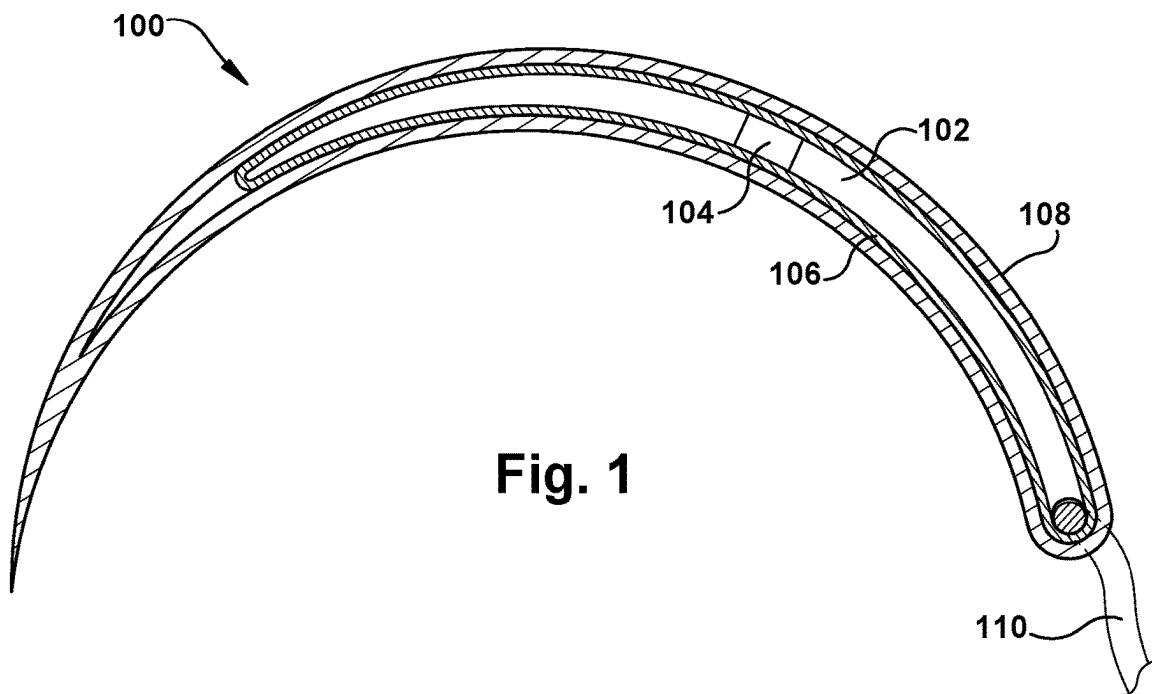
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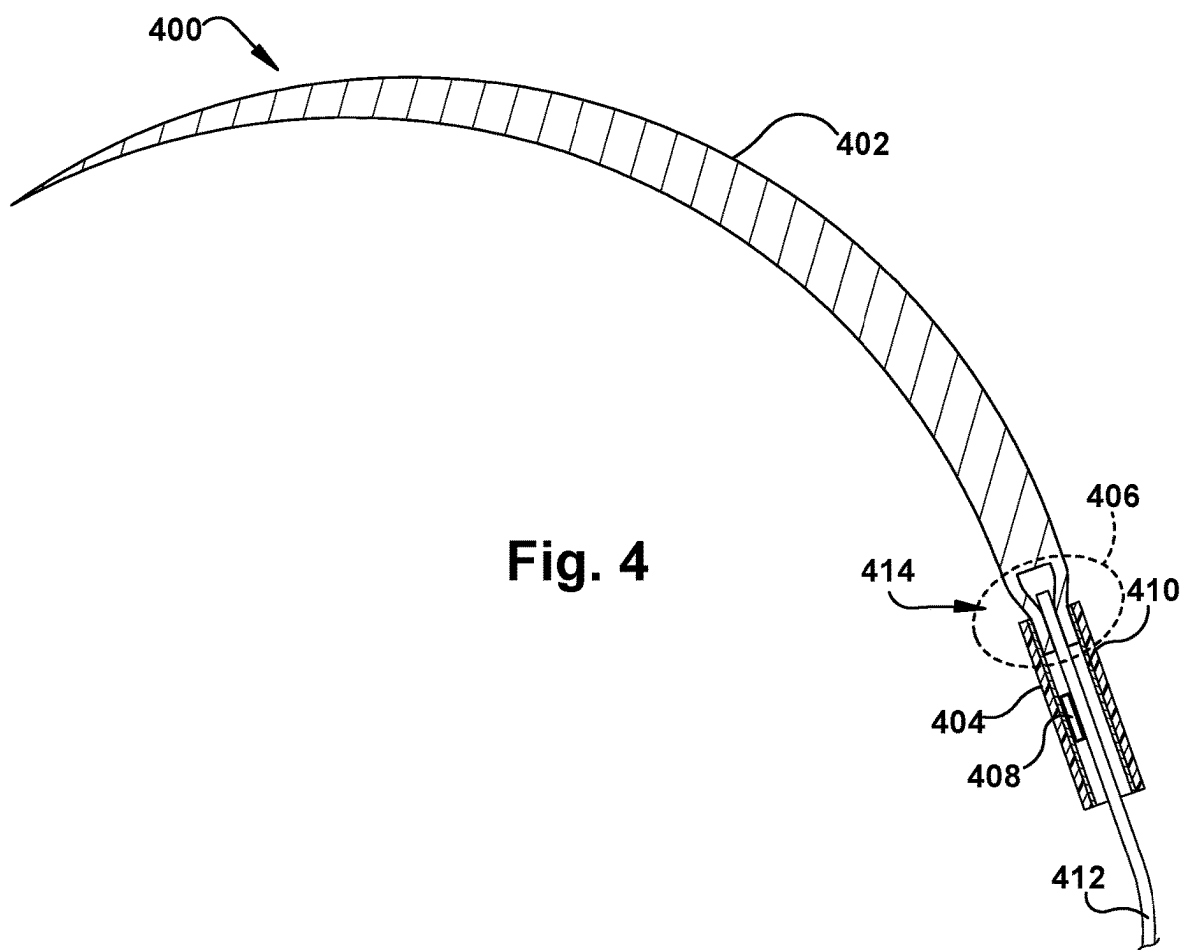
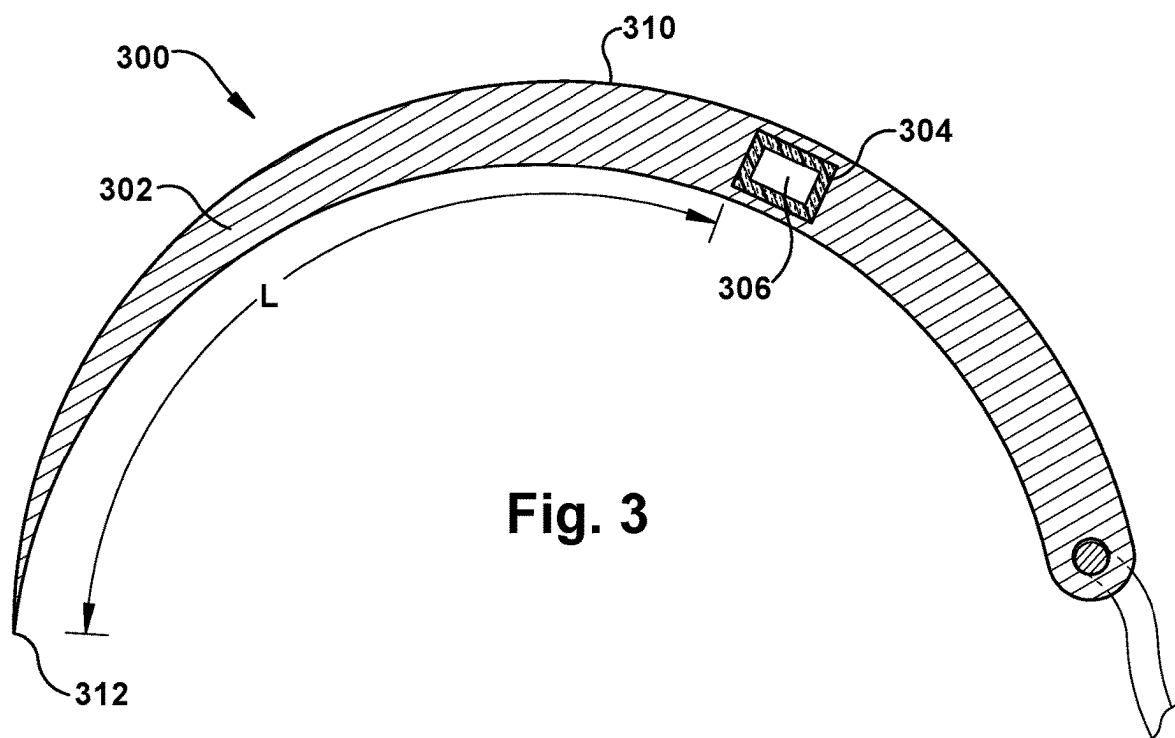
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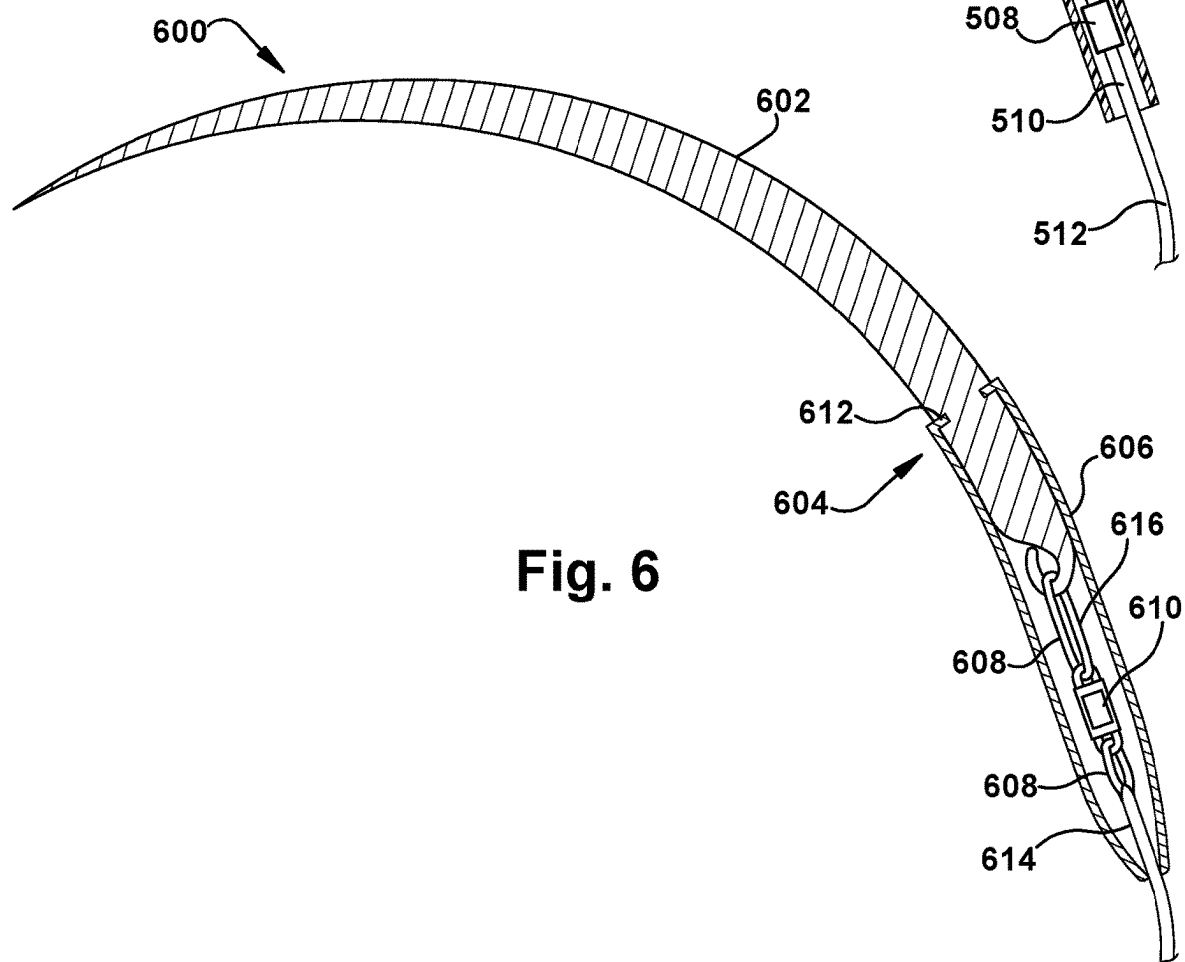
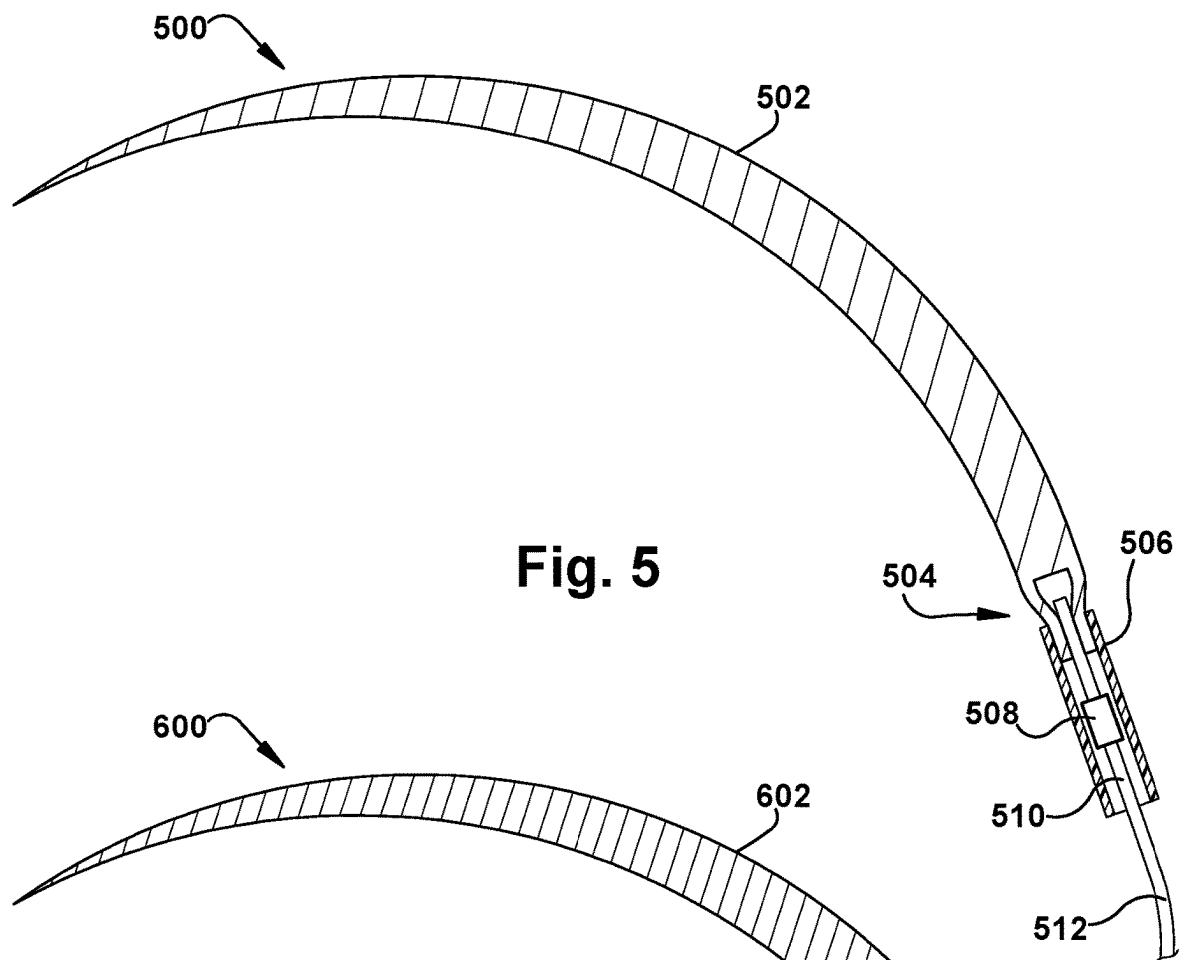
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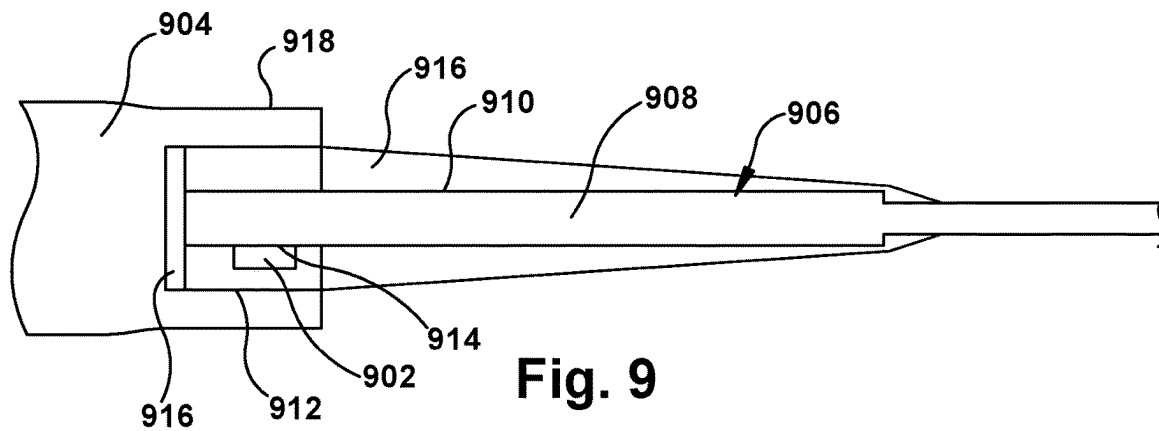
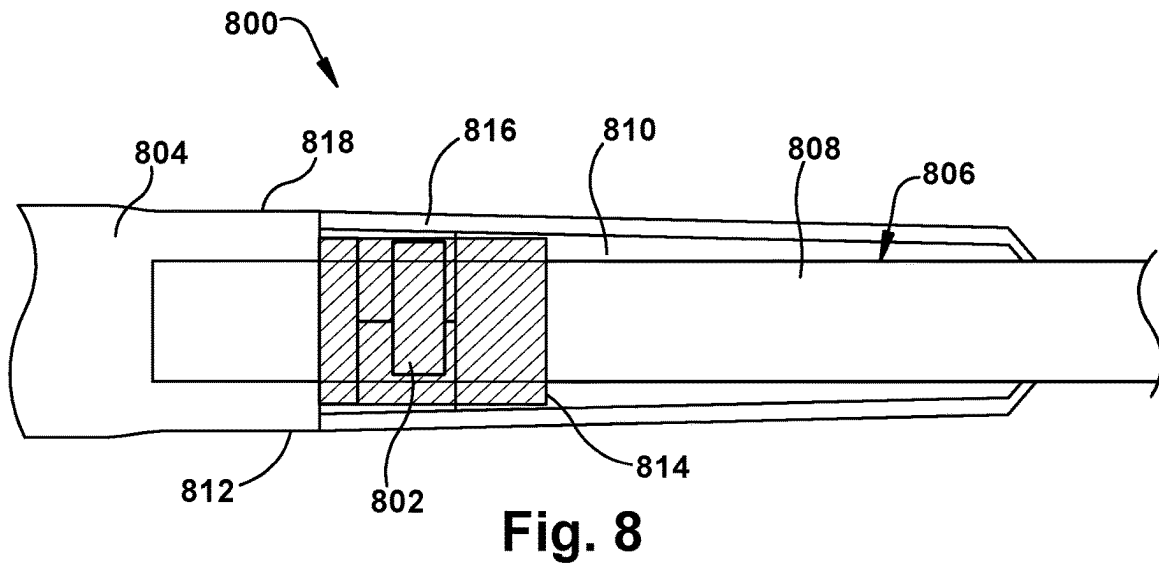
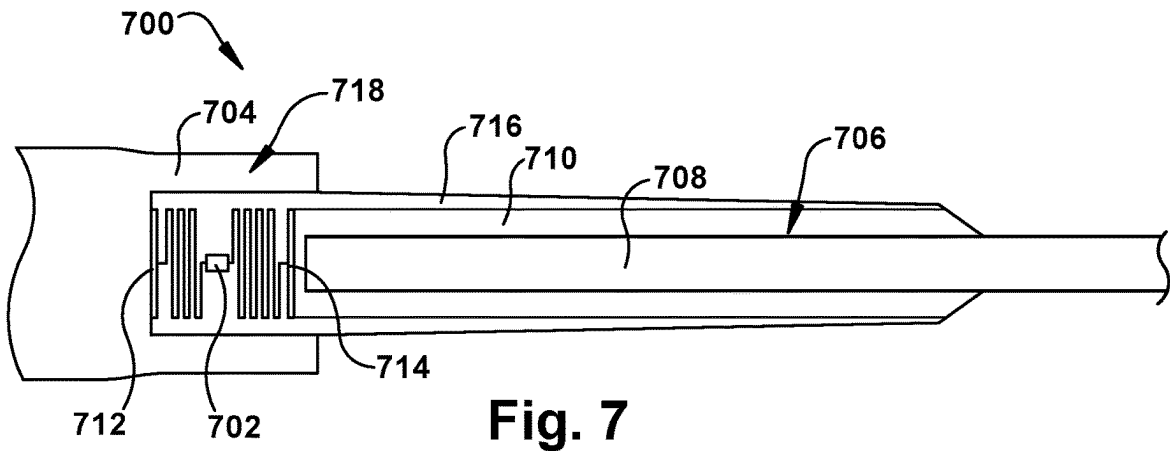
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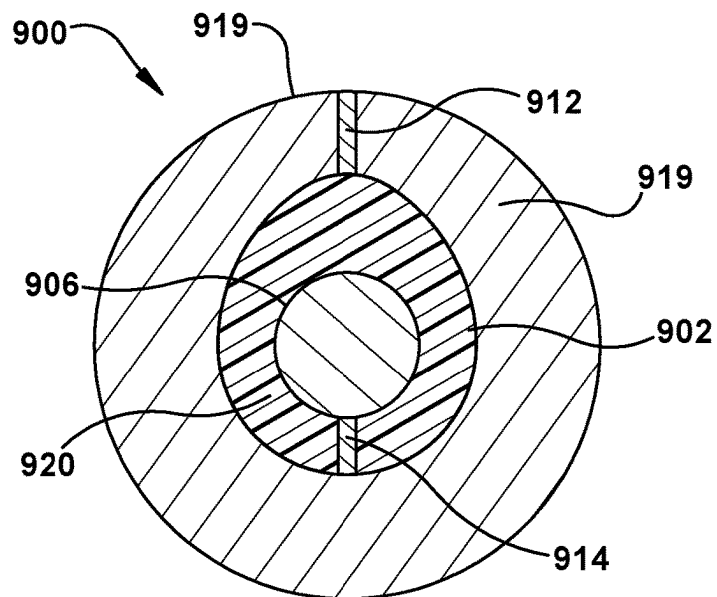


Fig. 10

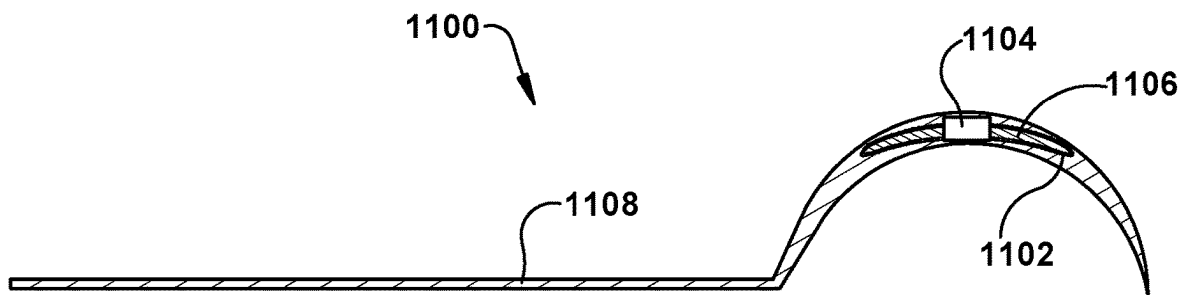
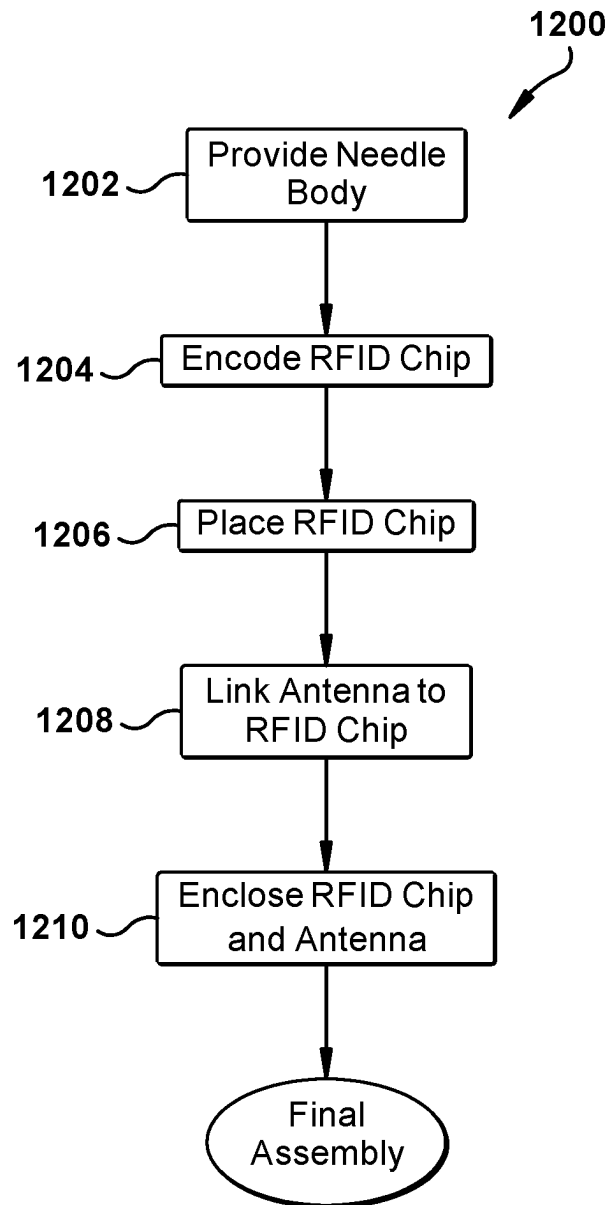
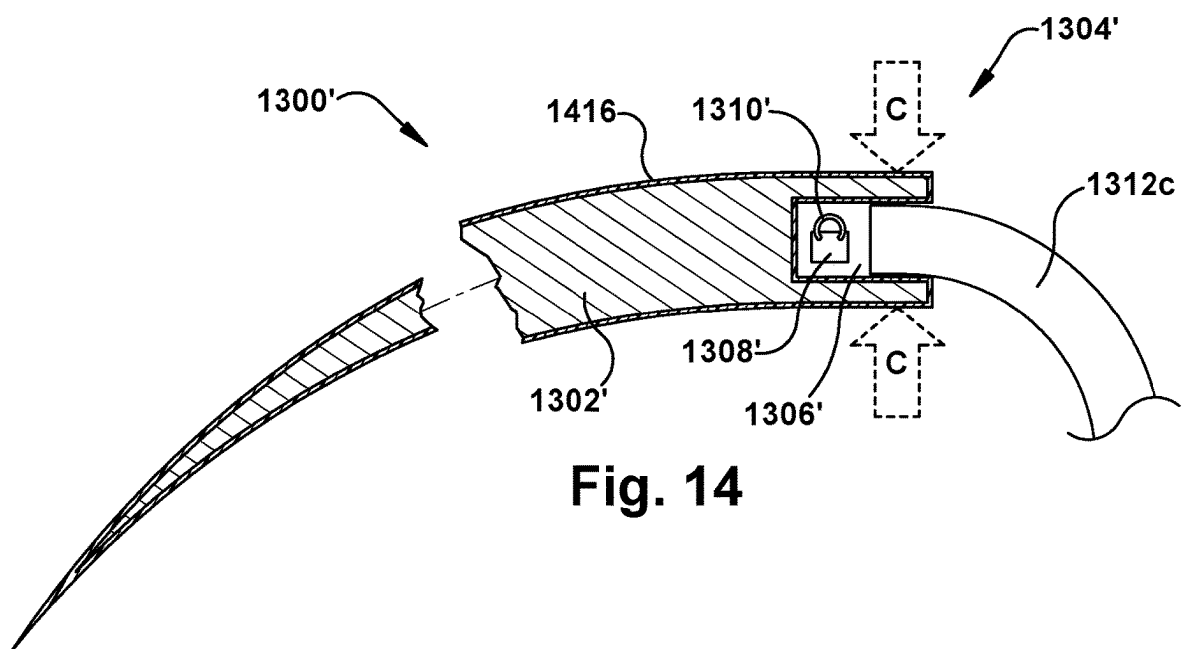
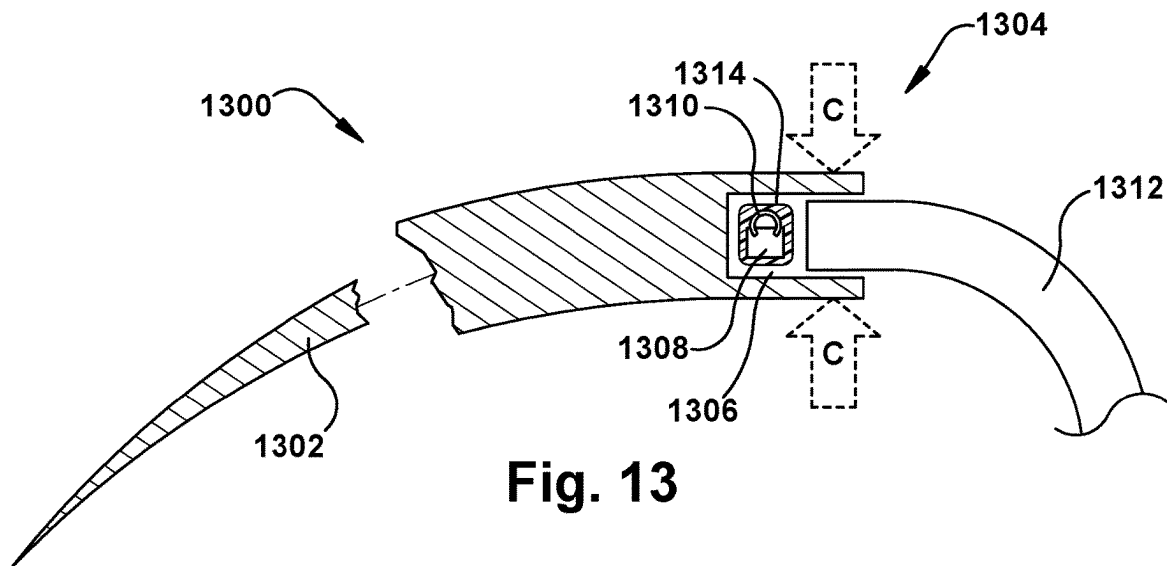


Fig. 11

**Fig. 12**



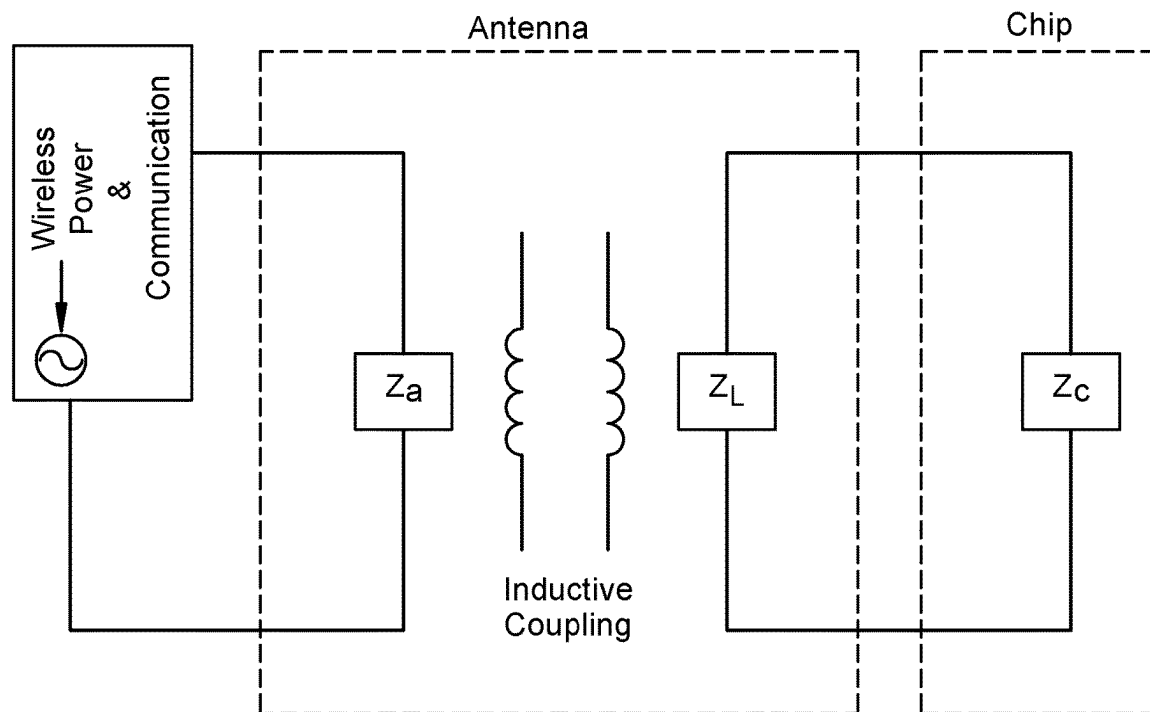


Fig. 15

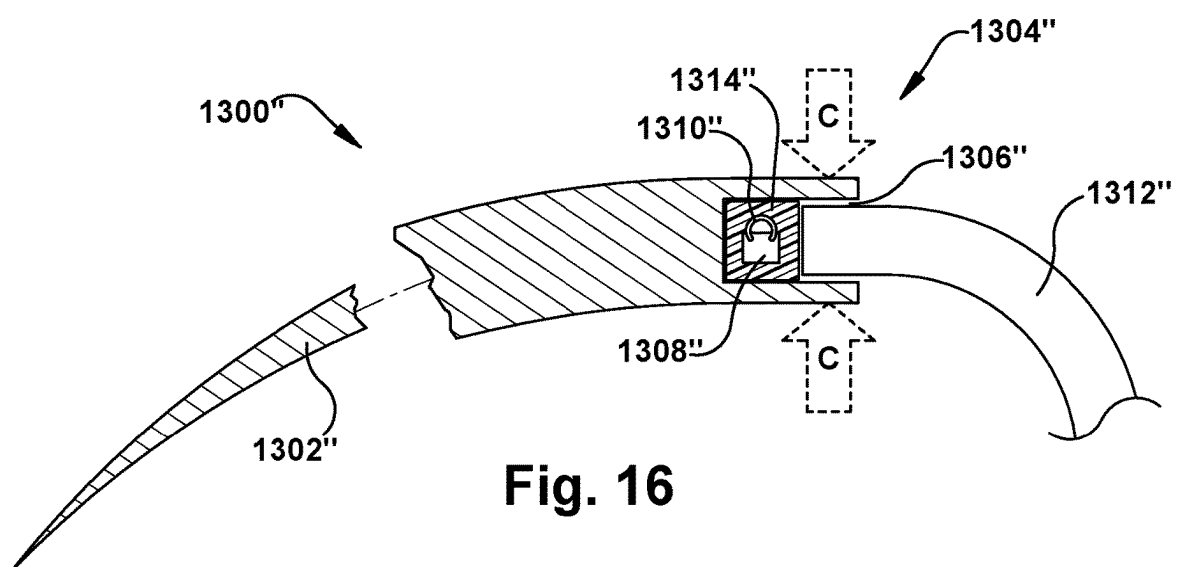
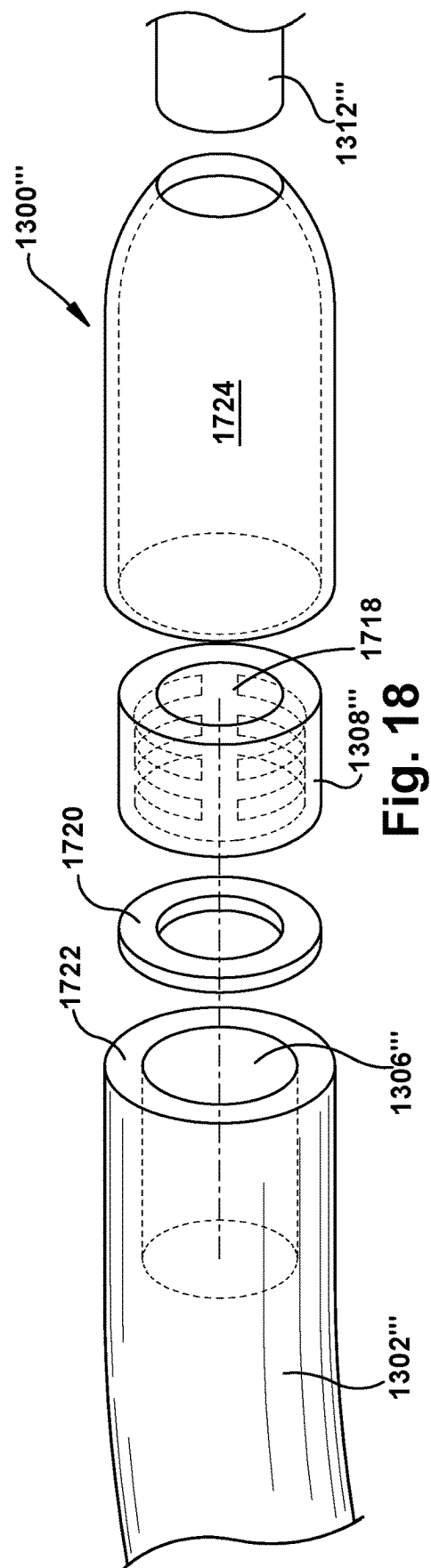
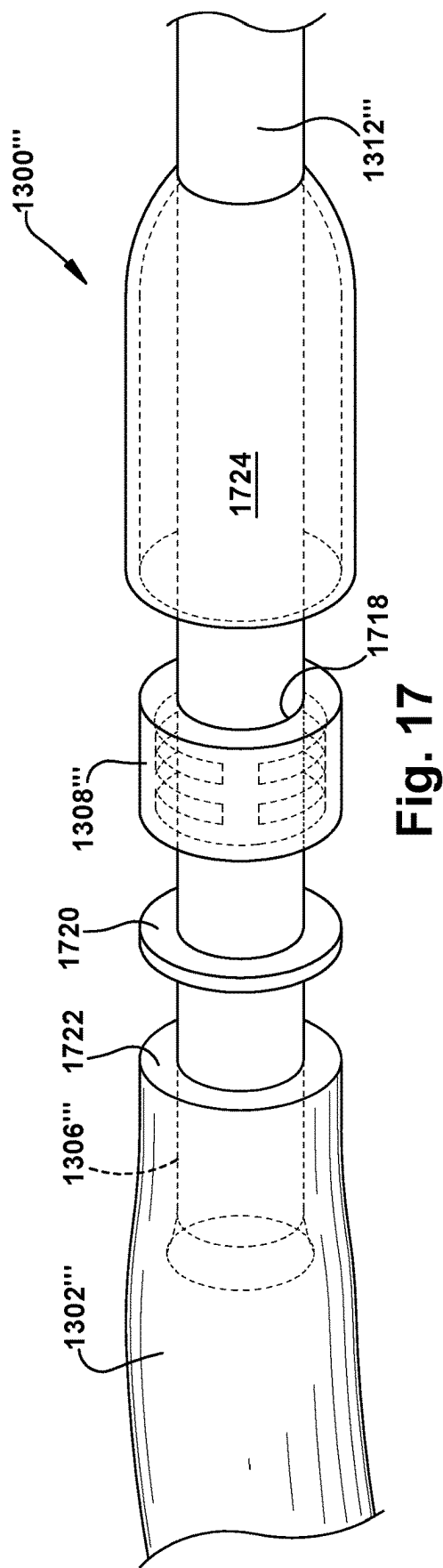
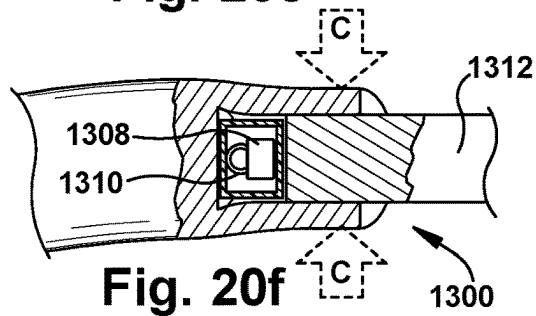
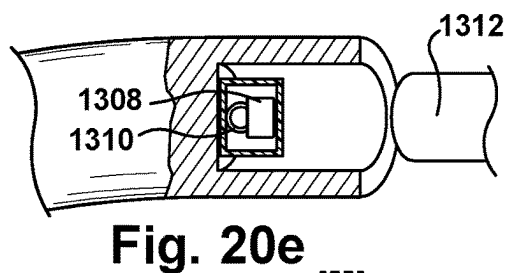
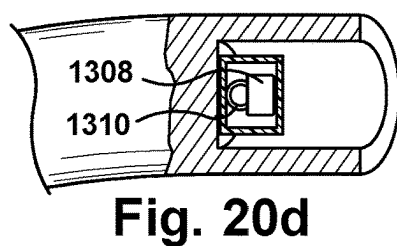
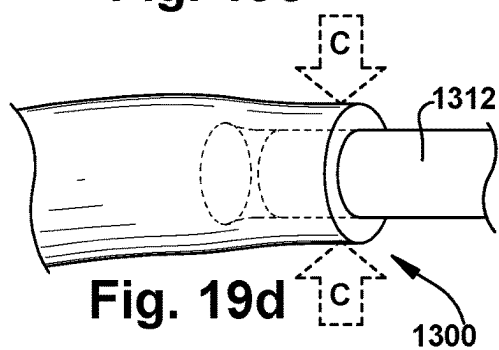
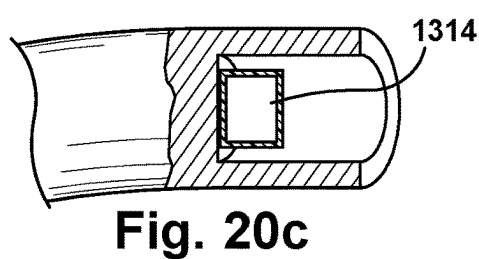
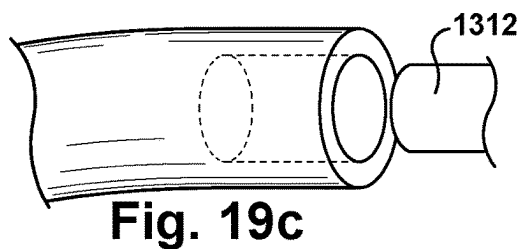
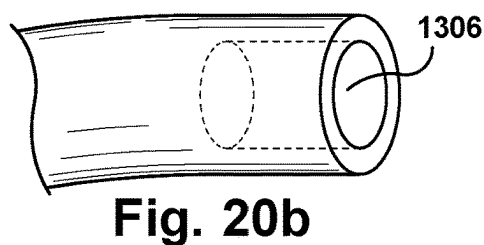
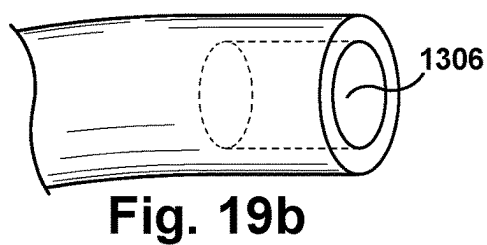
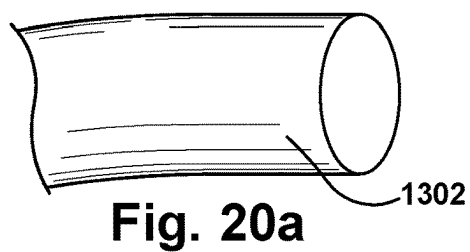
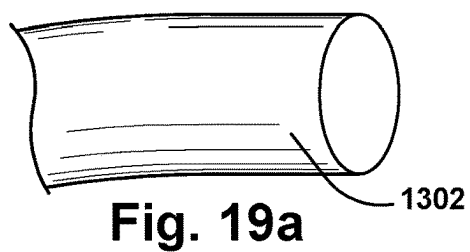
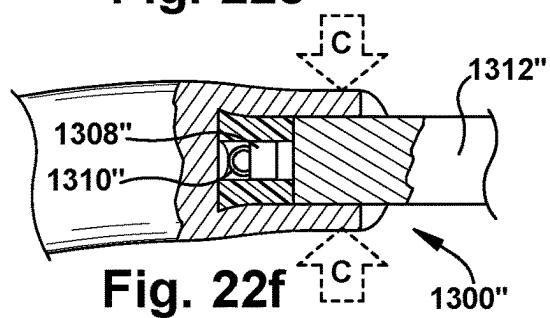
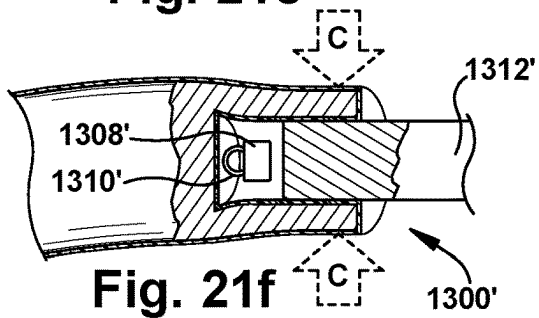
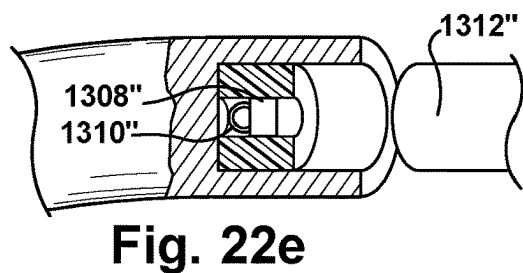
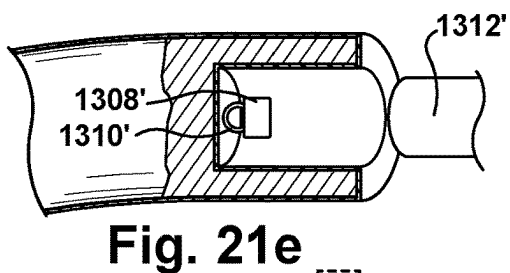
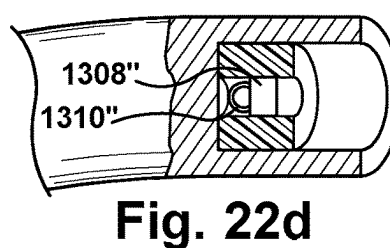
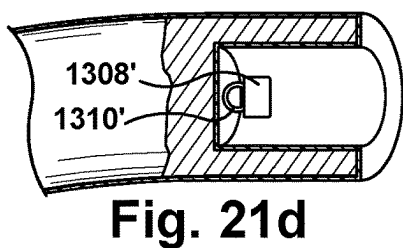
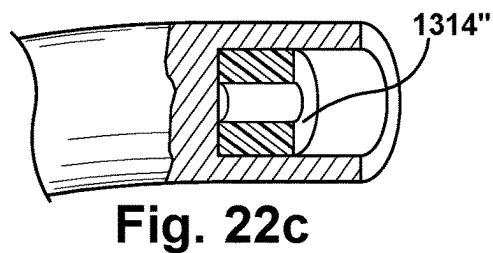
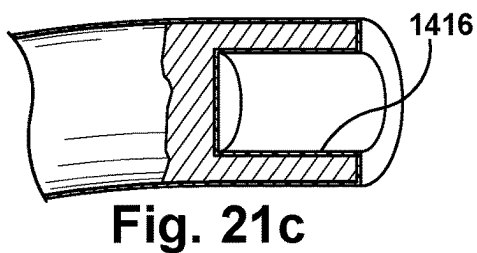
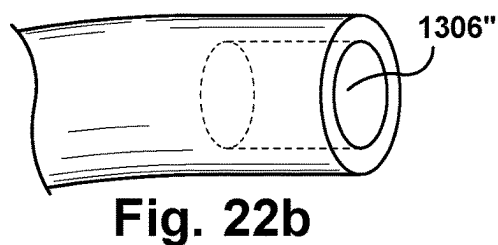
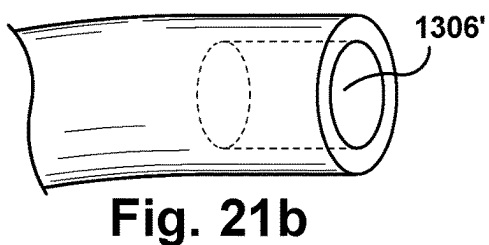
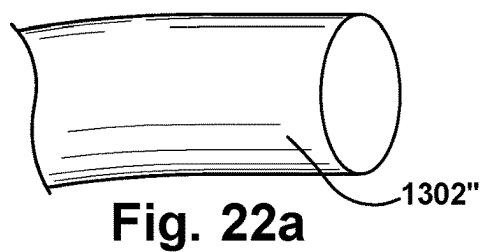
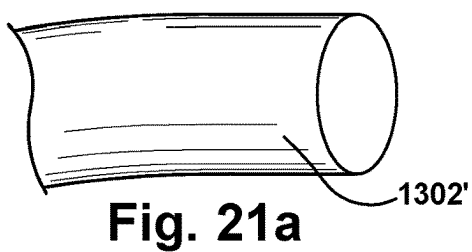


Fig. 16







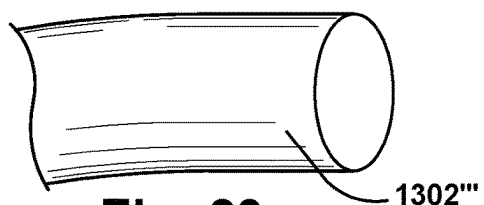


Fig. 23a

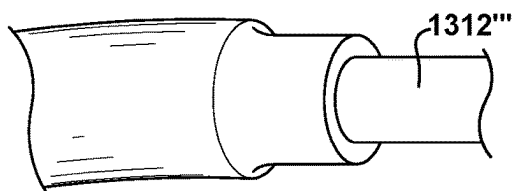


Fig. 23e

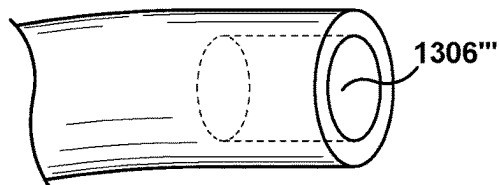


Fig. 23b

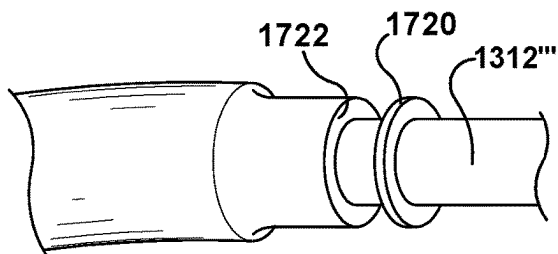


Fig. 23f

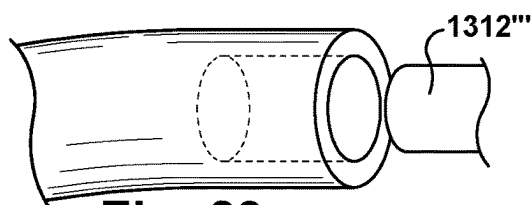


Fig. 23c

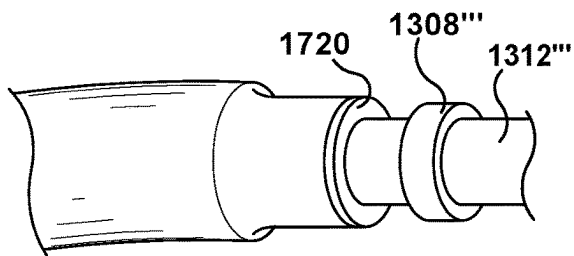


Fig. 23g

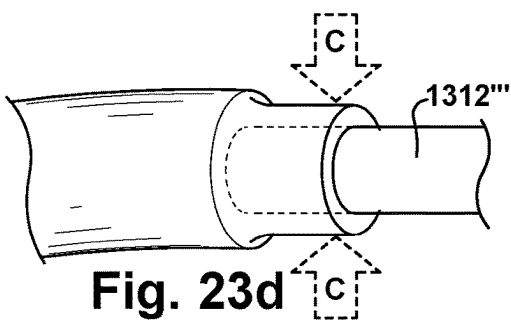


Fig. 23d

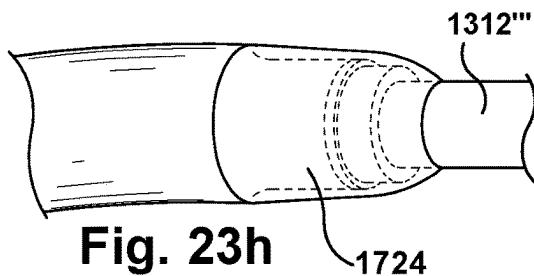


Fig. 23h

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WIRELESS-ENABLED SURGICAL SUTURE NEEDLE

RELATED APPLICATION

This application is a national phase application of and claims priority from PCT International Patent Application PCT/US2019/032103, filed May 14, 2019, which claims priority from U.S. Provisional Application No. 62/671,084, filed 14 May 2018. The subject matter of each of the aforementioned applications is incorporated herein by reference in its entirety.

TECHNICAL FIELD

This disclosure relates to an apparatus and method of manufacture of a surgical suture needle configured for wireless communication and, more particularly, to a wireless and/or RFID-enabled surgical suture needle.

BACKGROUND

Surgical suture needles are commonly used to hold separated body tissue together. The body tissue can be separated due to surgery or injury. During surgery, the number and location of all surgical devices (including surgical needles) within the surgical field should be regularly counted to ensure patient safety and to prevent retention of foreign objects within the patient. Protocols for maintaining accurate surgical counts remain predominantly manual processes and prone to human error.

Tracking systems based on high resolution video and wireless communication technologies have been proposed to improve the reliability and efficiency of this surgical count. Known tracking systems are capable of counting and locating larger objects, such as absorptive lap pads and hand-held surgical instruments, but are not capable of counting and locating smaller objects such as suture needles. The primary barriers to applying wireless tracking technologies to suture needles is their small size and traditional composition of metal which may interfere with signal transmission. However, missing or miscounted suture needles are the most common reason for an incorrect surgical count. Unlike larger instruments, needles can be difficult or impossible to visualize with x-rays that are performed to “rule out” the presence of a retained object. Thus, the prevalence of unresolved surgical counts involving a missing surgical needle remains high.

SUMMARY

In an aspect, a suture needle apparatus is described. The suture needle apparatus includes a needle body and a radio-frequency identification (“RFID”) chip disposed proximate to the needle body. The RFID chip is encoded with an identifying information associated with the suture needle apparatus. The RFID chip includes an electromagnetic coupling element. A suture thread is operatively coupled to the needle body. At least one of the needle body and the suture thread is an antenna selectively electromagnetically coupled to the RFID chip and, when coupled, is configured to wirelessly communicate the identifying information responsive to radio-frequency interrogation of the suture needle apparatus.

In an aspect, a method of providing a suture needle apparatus is described. A needle body and a radio-frequency identification (“RFID”) chip including an electromagnetic

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coupling element are provided. The RFID chip is encoded with information associated with the suture needle apparatus. The RFID chip is disposed proximate to the needle body. A suture thread is operatively coupled to the needle body. At least one of the needle body and the suture thread is selectively electromagnetically coupled to the RFID chip as an antenna. When the antenna and the RFID chip are electromagnetically coupled, the identifying information is wirelessly communicated responsive to radio-frequency interrogation of the suture needle apparatus.

BRIEF DESCRIPTION OF THE DRAWINGS

For a better understanding, reference may be made to the accompanying drawings, in which:

FIG. 1 is a schematic cross-sectional view of an example RFID-enabled surgical suture needle according to a first aspect;

FIG. 2 is a schematic cross-sectional view of an example RFID-enabled surgical suture needle according to a second aspect;

FIG. 3 is a schematic cross-sectional view of an example RFID-enabled surgical suture needle according to a third aspect;

FIG. 4 is a schematic cross-sectional view of an example RFID-enabled surgical suture needle according to a fourth aspect;

FIG. 5 is a schematic cross-sectional view of an example RFID-enabled surgical suture needle according to a fifth aspect;

FIG. 6 is a schematic cross-sectional view of an example RFID-enabled surgical suture needle according to a sixth aspect;

FIG. 7 is a schematic cross-sectional view of a portion of an example RFID-enabled surgical suture needle according to a seventh aspect;

FIG. 8 is a schematic cross-sectional view of a portion of an example RFID-enabled surgical suture needle according to an eighth aspect; and

FIG. 9 is a schematic cross-sectional view of a portion of an example RFID-enabled surgical suture needle according to a ninth aspect.

FIG. 10 is a schematic cross-sectional view of a portion of an example RFID-enabled surgical suture needle according to the ninth aspect.

FIG. 11 is a schematic cross-sectional view of a portion of an example RFID-enabled surgical suture needle according to a tenth aspect.

FIG. 12 is a flow chart illustrating an example method for manufacturing a suture needle apparatus.

FIG. 13 is a schematic cross-sectional view of a portion of an example RFID-enabled surgical suture needle according to an eleventh aspect.

FIG. 14 is a schematic cross-sectional view of a portion of an example RFID-enabled surgical suture needle according to a twelfth aspect.

FIG. 15 is an example electric circuit diagram of at least one aspect of the RFID-enabled surgical suture needle.

FIG. 16 is a schematic cross-sectional view of a portion of an example RFID-enabled surgical suture needle according to a thirteenth aspect.

FIG. 17 is a schematic cross-sectional view of a portion of an example RFID-enabled surgical suture needle according to a fourteenth aspect.

FIG. 18 is an exploded view of the RFID-enabled surgical suture needle of FIG. 17.

FIGS. 19a-d schematically depict an example assembly sequence of a prior art surgical suture needle.

FIGS. 20a-f schematically depict an example assembly sequence of the RFID-enabled surgical suture needle of FIG. 13.

FIGS. 21a-f schematically depict an example assembly sequence of the RFID-enabled surgical suture needle of FIG. 14.

FIGS. 22a-f schematically depict an example assembly sequence of the RFID-enabled surgical suture needle of FIG. 16.

FIGS. 23a-h schematically depict an example assembly sequence of the RFID-enabled surgical suture needle of FIG. 17.

DETAILED DESCRIPTION

The invention comprises, consists of, or consists essentially of the following features, in any combination.

Example surgical suture needles are described herein that address the barriers described and enable surgical suture needles to be equipped with wireless communication devices, which are described herein as radio-frequency identification (“RFID”) wireless chips that house identifying information and antennas that permit wireless communication, thereby enabling labeling, counting, and location and time tracking of surgical suture needles. Although references herein are made specifically to surgical suture needles, the example designs may similarly be applied to other surgical tools such as scalpel blade, cautery blade, forceps, clamps, retractors, angiocatheter, catheter, hypodermic needle, biopsy needle, port, probe, sponge, cannula, balloon, capsule, wire, stimulator, sensor, stent, coil, bead, particle, plug, clip, staples, marker or any other desired surgical or interventional instrument or tool.

Although the examples described herein make specific reference to RFID technology, the example aspects may similarly be implemented using other suitable wireless communication technology such as, but not limited to, UHF, HF, NFC, and/or any suitable battery-less technology(ies) spanning across broad radio frequency spectrums.

As part of a system for enabling the surgical sharp count, the examples described herein can be used to inventory, count, track, and/or locate suture needles inside and outside of the operative field. Furthermore, the example aspects can facilitate augmented reality for minimally invasive surgical approaches as well as fully automated robotic surgeries by permitting the identification and localization of the suture needle. In addition, the example aspects shown and described herein could be used to assist with identification and/or localization of one or more items which are intended to remain within the patient’s body over a longer period of time than the surgical procedure in which they are implanted.

A surgical suture needle for enabling labeling, counting, and/or location and time tracking includes a needle body and an RFID chip positioned proximate to the needle body. The RFID chip is encoded with identifying information associated with the suture needle, such as suture size, type, lot, unique identifier, expiration date, and other manufacturing information. The surgical suture needle includes an antenna for wirelessly communicating the identifying information. As will be described, the antenna may assume various forms, depending on the positioning of the RFID chip with respect to the needle body. Certain aspects of the surgical needle may also include a sheath made of RF lucent material that at least partially encloses and protects the RFID chip

and the antenna. Because one limiting factor for RFID communication efficiency and range is an effective size of the antenna, it may be desirable, in some use environments, to have the antenna be a rather large structure of the entire surgical needle system.

A “sheath,” as used herein is an outer covering for protecting the RFID chip from abrasion and damage which may result during use of the suture needle, during a sewing operation for example. The sheath is configured such that it will not interfere with the function of the suture needle. The sheath is also configured such that it allows RF energy to pass for the purpose of wireless communication.

An “inner core” as used herein is an RF lucent structure conforming to a shape of a suture needle.

Although the example suture needles described herein reference a single RFID chip, it should be appreciated that an example suture needle may benefit from including multiple RFID chips. For example, by including multiple RFID chips, a location of an example suture needle may be more easily identified using triangulation.

FIG. 1 illustrates a first aspect of an example RFID-enabled surgical suture needle (“suture needle”) 100. The suture needle 100 includes an inner core 102 or needle body that maintains the form of the suture needle 100. The suture needle 100 further includes an RFID chip 104 embedded in the inner core 102. The RFID chip 104 is encoded with a unique identifier associated with a particular suture needle 100. The suture needle 100 further includes an antenna 106 that is linked to the RFID chip 104. The RFID chip 104 is also configured to wirelessly receive and transmit identifying information. The antenna 106 is located within an outer sheath 108. The antenna spans the surface of the inner core 102 in an orientation which may be modified to optimize various radiofrequency properties. In other words, the RFID chip 104 and antenna 106 are coextensive with the outer sheath 108. The antenna 106 may optionally be electrically isolated from the inner core 102 with an insulating film.

The outer sheath 108 at least partially encloses the antenna 106 and the RFID chip 104 and protects the antenna 106 and the RFID chip 104 from external trauma and from other trauma, for example, related to pressure from a needle driver. It should be appreciated that, in one example, the inner core 102 in combination with the embedded RFID chip 104 and the enveloping antenna 106, enclosed by the outer sheath 108, provides for a form factor similar to a form factor of a known suture needle and is therefore familiar to a surgeon. It should be further appreciated that, in other examples, the inner core 102 in combination with the embedded RFID chip 104 and the enveloping antenna 106, enclosed by the outer sheath 108, may provide for a different form factor.

A suture 110 extends from the end of the suture needle 100. It should be appreciated that suture needle 100 (as well as the other example suture needles described herein) may include an eye loop (not shown) for selectively receiving the suture 110 there through, or could instead be swaged—that is, manufactured with the suture 110 attached to the needle—as desired for a particular use environment.

The suture needle 100 may be at least partially constructed of radiofrequency (“RF”) lucent material in order to allow radio waves to pass through without being blocked and without experiencing a loss in signal quality. In one example, both the inner core 102 and the outer sheath 108 may be constructed of the RF lucent material. In one example, the inner core 102 may be constructed of a traditional steel or alloy while the outer sheath 108 is constructed of RF lucent material. In such an example, an

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insulating film (not shown) may separate the inner core **102** from the RFID chip **104** and the antenna **106**. Although certain types of a material may be RF lucent with respect to certain frequencies, the same material may be RF-opaque and block other radio frequencies from passing. Thus, the selection of the RF lucent material may vary by application.

In one example, the RF lucent material may include, for example, Kevlar® aramid fiber (available from DuPont Protection Technologies of Richmond, VA), a carbon fiber reinforced polymer (“CFRP”), and/or any other desired polymer. In one example, CFRP may be electrically conductive, a property which may be advantageously used in the present aspects. In one example, the RF lucent material may include an organic material such as bamboo. In one example, the RF lucent material may include obsidian. It should be appreciated that the above-listed materials are described for illustrative purposes and that any other suitable material that allows for radio frequencies to pass without significantly impacting the quality of the signal may be used as a RF lucent material in the example aspects. It should be further appreciated that the materials described might not yet be used in FDA-approved devices and may require FDA testing/approval before being utilized in the example aspects. Although the example RF lucent materials have been described with reference to the suture needle **100** illustrated in FIG. 1, the example RF lucent materials may similarly be used in constructing one or more of the additional example suture needles described below.

In one example, the suture **110** may be made of conductive fiber or other material or may be coated in a conductive material. In such an example, the suture **110** is electrically isolated from the outer sheath **108** but is electrically connected to the antenna **106** embedded in the inner core **102**, thus forming a second arm or an extension of the antenna **106**. In one example, the suture **110** electrically connected to the antenna **106** extends the overall length of the antenna **106**. It should be appreciated that this increased antenna **106** length increases the amount of power available for RF communications. In one example, when extended read range is desired, a non-conductive RF lucent coating is used to bond, protect, or enhance, the electromagnetic properties of the suture needle **100**.

FIG. 2 illustrates a second aspect of an RFID-enabled surgical suture needle (“suture needle”) **200**. The suture needle **200** of the second aspect, shown in FIG. 2, includes an RFID chip **204** and an antenna **206** embedded in or proximate to an inner core **202**, and enveloped by an outer sheath **208**. This comprises the body **214** of the suture needle **200**. The inner core **202** and outer sheath **208** of the body **214**, as previously described, is constructed of RF lucent material. In this example aspect, however, the suture needle **200** is formed of a hybrid construct. In particular, the suture needle **200** includes a load bearing tip **210** constructed of stainless steel (or other suitable material, such as, but not limited to, a steel alloy). The tip **210**, which includes a point **212**, extends along the body of the suture needle **200** for a length “L”. The tip **210** is fused to the body **214** of the suture needle **200** at a bonding point **216**. Bonding of the tip **210** to the body **214** may performed using any suitable techniques such as friction bonding, for example. It should be appreciated that the hybrid design of the suture needle **200** will provide a similar suturing experience and tactile feedback as compared to a known suture needle **100** while incorporating RFID technology.

FIG. 3 illustrates a third aspect of an example suture needle **300**. A body **302** of the suture needle **300** may be constructed primarily of stainless steel (or other suitable

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conductive material, such as, but not limited to, a steel alloy). The suture needle **300** includes a relatively small gap or recess **304** cut into the body **302**. The suture needle **300** further includes an RFID chip **306** spanning the recess **304**. The recess **304** may be a square shape as illustrated or may be any other suitable shape for housing the RFID chip **306**. The conductive needle body **302** is connected directly to the RFID chip **306** at the recess **304** and therefore functions as an antenna, thereby eliminating the need for additional hardware or components. The recess **304** contains an RF lucent non-conductive material that serves the dual purpose of A) structurally supporting the chip B) electrically isolating it from the metal underneath. In one example, the recess **304** as well as the RFID chip **306** are together coated with RF lucent material to prevent damage to the chip due to abrasive forces. It should be appreciated that the location of the recess **304** in the body **302** of the suture needle **300** will be at some length “L” from the point **312** which may be selected such that mechanical stress and pressure on the RFID chip **306** resulting from the suture needle **300** driver and tissue manipulation is minimized. In one example, the RFID chip **306** is potted into the recess **304** with epoxy. This protects the RFID chip **306**, helps reduce friction, fosters mechanical integrity and support, and provides for a traditional needle form and shape.

In one example, the suture needle **300** further includes an outer sheath **310** that further protects the RFID chip **306** from external trauma and from other trauma related to pressure from needle driver.

It should be appreciated that, in the example suture needle **300**, the RFID chip **306** is slim enough such that it can be disposed inside the recess **304**. In one example, the RFID chip **306** is exposed and does not require a covering. In one example, the RFID chip **306** may be engraved onto the surface of the suture needle **300**.

FIG. 4 illustrates a fourth aspect of an example RFID-enabled surgical suture needle (“Suture needle”) **400**. The body **402** of the suture needle **400** is constructed primarily of stainless steel (or other suitable conductive material, such as, but not limited to, a steel alloy). The suture needle **400** includes a sheath **404** bonded to the suture needle **400** at bonding zone **406** at the swage end **414**. In other words, the sheath **404** acts as a cap that goes on the swage end **414** of the suture needle **400**. The sheath **404** includes an RFID chip **408** encoded with a unique identifier associated with the suture needle **400**. An antenna is comprised of the metal needle body **402** and a length of conductive material **410** within the sheath which extends away from the bonding zone **406**. The sheath extends from the needle body so as to maintain the silhouette of the suture needle **400** and to provide the surgeon with the familiar feel of a traditional suture needle. The sheath **404** is configured to allow for a suture **412** to extend out from the swage end, through the bonding zone **406**, and out through the sheath **404**. In one example, the sheath **404** may be at least partially constructed of RF lucent material as described.

FIG. 5 illustrates a fifth aspect of an example RFID-enabled surgical suture needle (“suture needle”) **500**. The suture needle **500** includes a body **502** constructed primarily of stainless steel (or other suitable material, such as, but not limited to, a steel alloy). At the swage end **504**, the suture needle **500** includes a sheath **506** or a cap with an embedded RFID chip **508** encoded with a unique identifier associated with the suture needle **500**. The metallic body **502** of the suture needle **500** serves as an antenna for wireless communication. In contrast to the example suture needle **400** illustrated in FIG. 4, however, portions of the metallic body

502 of the suture needle **500** do not extend and overlap the sheath **506**. Rather, the suture needle **500** includes a conductive suture **510** that extends from the swage and through the sheath **506**. The conductive suture **510** is at least partially made of a conductive material, thereby serving as an extension of the metallic body **502** as well as a trailing antenna. In one example, a non-conductive thread **512** extends out beyond the sheath **506** and thus resembles a traditional suture of a traditional suture needle **100**. Thus, a portion of conductive suture **510** within the sheath **506** serves as a second arm of the antenna **510**. In one example, to further facilitate wireless communication, at least a portion of the sheath **506** contains or is coated with an electrically conductive material.

FIG. 6 illustrates a sixth aspect of an example RFID-enabled surgical suture needle ("suture needle") **600**. The suture needle **600** includes a body **602** constructed primarily of stainless steel (or other suitable material, such as, but not limited to, a steel alloy). At or near the swage end **604**, the suture needle **600** includes a sheath **606** constructed of RF lucent material.

In one example, the sheath **606** may be coupled to the metallic body **602** at the swage end **604** using a clamp **612**. It should be appreciated that other suitable bonding techniques may be used to couple the sheath **606** to the metallic body **602**. Within the sheath **606**, the suture needle **600** includes a conductive suture **608** coupled to the metallic body **602** at the swage end **604**. The suture needle **600** further includes an RFID chip **610** encoded with a unique identifier associated with the suture needle **600**. The RFID chip **610** is positioned such that it spans the metallic body **602** and the conductive suture **608**, which form two arms of an antenna. It should be appreciated that, although the conductive suture **608** is in close proximity to the metallic body **602**, the lagging end **614** of the conductive suture **608** remains electrically isolated from the metallic body **602** while the shorter end **616** of the conductive suture **608** is electrically connected to the metallic body **602**.

The RFID chip **610** may be oriented in a variety of ways to facilitate contact as described. For example, as illustrated in a seventh aspect of an example RFID-enabled surgical suture needle ("suture needle") **700** in FIG. 7, the suture needle **700** includes an RFID chip **702** positioned axially between a metallic body **704** and a conductive suture **706**. The conductive suture **706** includes a standard suture **708** covered by a conductive coating **710**. A chip-body contact **712**, shown here as a first end plate, connects the metallic body **704** to the RFID chip **702**. Thus, the metallic body **704** and the conductive suture **706** together form the two arms of an antenna. A chip-suture contact **714**, shown here as a second end plate, connects the conductive suture **706** to the RFID chip **702**. The suture needle **700** includes an RF lucent sheath **716** constructed of RF lucent material that wraps around the conductive suture **706** and the RFID chip **702** and serves as an electrical insulator. In one example, a portion of the metallic body **704** may extend and partially overlap at least a portion of the RFID chip **702** and the RF lucent sheath **716** at the swage end **718**. The conductive suture **706**, however, is not completely overlapped by the metallic body **704** as shown in the Figures.

In an eighth aspect of an example RFID-enabled surgical suture needle ("suture needle") **800**, as illustrated in FIG. 8, the suture needle **800** includes an RFID chip **802** positioned between the metallic needle body **804** and a conductive suture **806**. In this aspect, the RFID chip **802** is a donut-like disk, threaded onto string. The conductive suture **806** is formed from a standard suture **808** with conductive coating

810. A chip-body contact **812**, shown as a first end plate, connects the metallic body **804** to the RFID chip **802**. A chip-thread contact **814**, shown as a second end plate, connects the conductive suture **806** to the RFID chip **802**. Thus, the RFID chip **802** is a two-port connection such that the metallic needle body **804** serves as one arm of an antenna across a first port and the conductive suture **806** serves as a second arm of the antenna across a second port. The suture needle **800** includes an RF lucent sheath **816** constructed of RF lucent material that wraps around the conductive suture **806** and the RFID chip **802** and serves as an electrical insulator. The RF lucent sheath **816** is joined to the metallic body **804** at a swage end **818**. In one example, a portion of the metallic body **804** may extend and partially overlap at least a portion of the conductive suture **806** at the swage end **818**. The conductive suture **806** is not completely overlapped by the metallic body **804** as shown in the Figures.

In a ninth aspect of an example RFID-enabled surgical suture needle ("suture needle") **900**, as illustrated in FIG. 9, the suture needle **900** includes an RFID chip **902** positioned proximate to a swage end **918** of a metallic body **904**. In this aspect, the RFID chip **902** is circumferentially wrapped around a conductive suture **906**. The conductive suture **906** includes a standard suture **908** covered by a conductive coating **910**. The chip-body contact **912**, electrically connects one end of the wrapped RFID chip **902** to the conductive metallic swage end **918**. The chip-suture contact **914**, electrically connects the other end of the wrapped RFID chip **902** to the conductive suture **906**. Thus, the metallic body **904** and the conductive suture **906** together form the two arms of an antenna.

As illustrated in the cross sectional view in FIG. 10, the chip-body contact **912** is supported by a circumferentially wrapped non-conductive outer layer **919**. The chip-suture contact **914** is also supported by a circumferentially wrapped non-conductive inner layer **920**. The inner and outer non-conductive layers **919** and **920** provide structural support to the chip-body contact **912**, the chip-suture contact **914**, and the RFID chip **902** and prevent inadvertent short circuiting of the chip **902**.

Referring again to FIG. 9, the suture needle **900** includes an RF lucent sheath **916** constructed of RF lucent material that wraps around the conductive suture **106** and serves as an electrical insulator. The RF lucent sheath **916** is joined to the metallic body **904** at a swage end **918**. In one example, a portion of the metallic body **904** may extend and partially overlap at least a portion of the RFID chip **902** and the conductive suture **906** at the swage end **918**. The conductive portion of the conductive suture **906**, however, is not completely overlapped by the metallic body **904**, as illustrated in FIG. 9.

In a tenth aspect of an example RFID-enabled surgical suture needle ("suture needle") **1100**, as illustrated in FIG. 11, a sheath **1102** that encloses the RFID chip **1104** and antenna **1106** is made from the same RF lucent material as the suture **1108**. Thus, the combination of the RFID chip **1104** and the antenna **1106** serve as the core of the suture **1108**. In one example, the RF lucent material may comprise an absorbable plastic such as polyglycolic acid. In another example, the RF lucent material may comprise a non-absorbable plastic such as polypropylene. Using the same material for both the suture and the sheath enclosing the suture may eliminate a need to work with and test additional materials, thereby simplifying the manufacturing process.

FIG. 12 a flow chart illustrating an example method **1200** for manufacturing a suture needle apparatus. At block **1202**, a needle body is provided. At block **1204**, an RFID chip is

encoded with information associated with the suture needle apparatus. At block **1206**, the RFID chip is disposed near the needle body. In one example, the RFID chip is disposed inside a recess of the needle body and the recess is filled with epoxy or another suitable binder/potting material. At block **1208**, an antenna is linked to the RFID chip. The antenna is configured to wirelessly communicate the identifying information. At block **1210**, the RFID chip and the antenna are at least partially enclosed with a sheath made of RF lucent material to form a final assembly.

FIG. **13** illustrates an eleventh aspect of an example RFID-enabled surgical suture needle ("suture needle") **1300**, which may include any of the features previously discussed with respect to other suture needles. The suture needle **1300** of FIG. **13** includes a body **1302** which is at least one of partially metallic, organic, plastic, and composite, such as by being constructed primarily of stainless steel (or other suitable material, such as, but not limited to, a steel alloy). At or near the swage end **1304**, the suture needle **1300** includes a cavity or recess **1306**. A radio-frequency identification ("RFID") chip **1308** disposed proximate to the needle body **1302**. The RFID chip **1308** is encoded with an identifying information, which may comprise at least one of suture size, type, lot, unique identifier, and expiration date, associated with the suture needle **1300** apparatus. The RFID chip **1308** includes an electromagnetic coupling element to facilitate inductive and/or capacitive connection of the RFID chip **1308** with the needle body **1302** and/or a suture thread, such as the inductive loop **1310**.

As shown in FIG. **13**, the RFID chip **1308** may be at least partially disposed inside the recess **1306** of the needle body **1302**. The recess **1306** may be configured to concurrently receive the RFID chip **1308** and at least a portion of a suture thread **1312** (shown here as a distal end of the suture thread **1312**).

The suture thread **1312** may be of any suitable type, such as, but not limited to, silk, metallic, gut, polyglycolic acid, polylactic acid, nylon, polypropylene, polyester, or the like. The suture thread **1312** may extend from the needle body **1302**, such that the suture needle **1300** may be used to draw the suture thread **1312** through a patient tissue in a known manner. The suture thread **1312** may be operatively coupled to the needle body **1302** in any desired manner. For example, the needle body may include a swaged end **1304** selectively coupled to the suture thread **1312**. As shown by arrows "C" in FIG. **13**, the swaged end **1304** may be crimped to directly connect the needle body **1302** and the suture thread **1312** (here, as previously mentioned, the distal end of the suture thread **1312**).

Also as shown in FIG. **13**, the RFID chip **1308** may be disposed axially between at least a portion of the needle body **1302** and the suture thread **1312**. In the aspect shown in FIG. **13**, this is accomplished by lining up the needle body **1302**, the RFID chip **1308** in the recess **1306**, and the suture thread **1312** longitudinally in a row as the structure is assembled.

At least one of the needle body **1302** and the suture thread **1312** may serve as an antenna selectively electromagnetically (e.g., inductively and/or capacitively) coupled to the RFID chip. When coupled, the antenna is configured to wirelessly communicate the identifying information responsive to radio-frequency interrogation of the suture needle **1300** apparatus. For example, in some use environments, at least a portion of the suture thread **1312** may comprises conductive material to provide a conductive suture thread **1312C**. The conductive suture thread **1312C** may be electrically isolated from the needle body **1302** (as shown and

will be discussed below with respect to FIG. **14**), and the conductive suture thread **1312C** is at least a portion of the antenna.

With reference again to FIG. **13**, the RFID chip **1308** may have a small inductive loop **1310** operatively coupled to it, when the electromagnetic coupling is done inductively. This inductive loop **1310** inductively couples to a conductive portion of the apparatus acting as an antenna, such as to a conductive portion (e.g., metal) of a conductive suture thread **1312C** and/or to the needle body **1302**. Therefore, the antenna (hereafter presumed to be a conductive suture thread **1312C**) coupled with the inductive loop **1310** acts to collect and supply power to the RFID chip **1308** to power it up, and the RFID chip **1308** can the backscatter the identifying information back to an RFID reader, assisted by the structure acting as an antenna. This electrical relationship is shown in FIG. **15**.

In order to facilitate this operation, it may be desirable, in some use environments, to electrically insulate the RFID chip **1308** (with its included inductive loop **1310**) from a chosen one of the conductive suture thread **1312C** and the needle body **1302** when the other one of the conductive suture thread **1312C** and the needle body **1302** is acting as the antenna. To that end, and using a conductive suture thread **1312C** as an example antenna, the RFID chip **1308** may be at least partially enclosed within a nonconductive shell **1314**, as shown in FIG. **13**. The nonconductive shell **1314**, when present, is configured to electrically isolate the RFID chip **1308** from the needle body **1302**. The nonconductive shell **1314** should, in most use environments, entirely three-dimensionally surround the RFID chip **1308** and can be made of any material, including, but not limited to, polymers, silicone, hot glue, parylene, glass, ceramic, plastic, paper or cellulose-based materials, and/or any other similarly nonconductive materials. Similarly, the RFID chip **1308** can be enclosed within the nonconductive shell **1314** in any desired manner including, but not limited to, placement and curing of a flowable hardening material around the RFID chip **1308** and assembly of a preconfigured nonconductive shell **1314** around the RFID chip **1308** (e.g., a multipiece shell or a shell having an opening to selectively admit the RFID chip **1308** therethrough).

FIG. **14** illustrates a twelfth aspect of a suture needle. The suture needle of FIG. **14** is similar to the suture needle of FIG. **13** and therefore, structures of FIG. **14** that are the same as or similar to those described with reference to FIG. **13** have the same reference numbers with the addition of a "prime" mark. Description of common elements and operation similar to those in the previously described eleventh aspect, or any other aspect herein, will not be repeated with respect to the twelfth aspect, but should instead be considered to be incorporated below by reference as appropriate.

In the suture needle **1300'** of FIG. **14**, the needle body **1302'** is at least partially coated with a nonconductive skin **1416** configured to electrically isolate the RFID chip **1308'** from the needle body **1302'**. This nonconductive skin **1416** may cover all of the needle body **1302'** or just a portion (e.g., an interior of the recess **1306'**) which is likely to come into unwanted otherwise-conductive contact with the RFID chip **1308'** and/or its included inductive loop **1310'**. The nonconductive skin **1416** may be homogeneously applied to the needle body **1302'** in any desired manner, including a hardening liquid coating (e.g., parylene) and/or a shrink-wrapped sheath, and may be made of any suitable material or combination of materials. It is contemplated that the nonconductive skin **1416** may have different thicknesses at different areas of the needle body **1302'**.

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FIG. 16 illustrates a thirteenth aspect of a suture needle. The suture needle of FIG. 16 is similar to the suture needle of FIG. 13 and therefore, structures of FIG. 16 that are the same as or similar to those described with reference to FIG. 13 have the same reference numbers with the addition of a double “prime” mark. Description of common elements and operation similar to those in the previously described eleventh aspect, or any other aspect herein, will not be repeated with respect to the thirteenth aspect, but should instead be considered to be incorporated below by reference as appropriate.

Suture needle 1300" shown in FIG. 16 is somewhat similar to that of FIG. 13, in that the needle body 1302" includes a recess 1306" for selectively receiving the RFID chip 1308", and the RFID chip 1308" and inductive loop 1310" are enclosed within a nonconductive shell 1314". However, in FIG. 16, the nonconductive shell 1314" closely conforms to at least one inner dimension of the recess 1306". Indeed, the nonconductive shell 1314" may be selectively frictionally engaged (i.e., press-fit) into the recess 1306", potted into the recess 1306", or otherwise placed into the configuration depicted in FIG. 16.

With this “plug-type” nonconductive shell 1314", the distal end of the suture thread 1312" can be butted up against, or even inserted slightly into, a proximal surface of the nonconductive shell 1314" before crimping. As opposed to the suture needle 1300 of FIG. 13, the suture needle 1300" of FIG. 16 may allow a fabricator to have greater control over the orientation and positioning of the RFID chip 1308" within the recess 1306", which may assist with standardization, quality control, and even operability of the suture needle 1300" apparatus.

FIGS. 17-18 illustrate a fourteenth aspect of a suture needle. The suture needle of FIGS. 17-18 is similar to the suture needle of FIG. 13 and therefore, structures of FIGS. 17-18 that are the same as or similar to those described with reference to FIG. 13 have the same reference numbers with the addition of a triple “prime” mark. Description of common elements and operation similar to those in the previously described eleventh aspect, or any other aspect herein, will not be repeated with respect to the fourteenth aspect, but should instead be considered to be incorporated below by reference as appropriate.

FIG. 18 is an exploded view of the suture needle 1300" apparatus of FIG. 17. This fourteenth aspect differs from those previously described at least because the RFID chip 1308" has an annular profile, including a center aperture 1718 configured to selectively receive at least a portion of the suture thread 1312". The needle body 1302" includes a recess 1306" for selectively concurrently receiving the annular RFID chip 1308" and a portion of the suture thread 1312". There may be at least one washer 1720 which mechanically and/or electrically isolates the annular RFID chip 1308" from other structures of the apparatus. For example, and as shown in FIG. 17, the washer 1720 (which may be an insulating washer) may be longitudinally interposed between a rim 1722 of the recess 1306" and the annular RFID chip 1308", with the suture thread 1312" extending entirely axially through both the washer 1720 and the RFID chip 1308" and the distal end of the suture thread 1312" being crimped into the recess 1306". In this case, the RFID chip 1308" is not located, or not entirely located, in the recess 1306" but is still proximate the needle body 1302".

In the aspect of FIGS. 17-18, the suture needle 1300" apparatus may include a hollow cap 1724 fitting about a circumference of the suture thread 1300" for preventing

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egress of the RFID chip 1308" from the suture needle 1300". Accordingly, the combination of the washer 1720 and cap 1724 may serve to electrically insulate the annular RFID chip 1308" from the needle body 1302" while holding the RFID chip 1308" in inductively and/or directly conductive contact (or any other electromagnetic contact) with the suture thread 1300". Use of the cap 1724 may help to smooth a mechanical transition between the needle body 1302" and suture thread 1300", and therefore assist in avoiding “catching” of the suture needle 1300" apparatus and/or provide a lower-profile transition area than if the RFID chip 1308" were to be located inside a recess 1306" of the needle body 1302".

A method of providing any of the suture needle apparatuses of at least FIGS. 13-18 may include providing a needle body 1300 and providing an RFID chip 1308 including an inductive loop 1310. (The numbering of FIG. 13 only is used here and below for simplicity, but does not preclude this description applying also or instead to the aspects of FIGS. 14-18. Also, the electromagnetic coupling is referenced herein as inductive coupling, for ease of description.) The RFID chip 1308 is encoded with information associated with the suture needle 1300 apparatus. The RFID chip 1308 is disposed proximate to the needle body 1302, which can include disposing at least a portion of the RFID chip 1308 inside a recess 1306 of the needle body 1302.

A suture thread 1312 is operatively coupled to the needle body 1302, such as by swaging the suture needle 1300 with the suture thread 1312. At least one of the needle body 1302 and the suture thread 1312 is inductively coupled to the RFID chip 1308 to serve as an antenna. When the suture thread 1312 is serving as the antenna, the RFID chip 1308 may be electrically isolated from the needle body 1302. For example, this isolation may be accomplished by at least partially enclosing the RFID chip 1308 within a nonconductive shell 1314—which could include closely conforming the nonconductive shell 1314 to at least one inner dimension of the recess 1306—and/or at least partially coating the needle body 1302 with a nonconductive skin 1416.

It is contemplated that the RFID chip 1308 could also or instead be provided with an annular profile including a center aperture 1718. In this case, at least a portion of the suture thread 1312 is extended through the center aperture 1718. Concurrently with the RFID chip 1308 surrounding at least a portion of the suture thread 1312, an other portion of the suture thread 1312 is placed into a recess 1306 of the needle body 1302. With a cap 1724 fit about a circumference of the suture thread 1312, egress of the RFID chip 1308 from the suture needle 1300 is provided. This is another way that the RFID chip 1308 can be electrically isolated from the needle body.

Regardless of how the electrical isolation occurs, though, when the antenna and the RFID chip 1308 are inductively coupled, the identifying information can be wirelessly communicated as desired responsive to radio-frequency interrogation of the suture needle 1300 apparatus.

FIGS. 19a-23h schematically depict assembly of, respectively, a prior art (non-RFID-equipped) suture needle (numbered analogously to FIG. 13), the RFID-enabled surgical suture needle of FIG. 13, the RFID-enabled surgical suture needle of FIG. 14, the RFID-enabled surgical suture needle of FIG. 16, and the RFID-enabled surgical suture needle of FIGS. 17-18. The various views of FIGS. 19a-23h can happen concurrently and/or sequentially, and in any order including an order different than the lower-case lettering would seem to indicate.

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In each of FIGS. 19a-23h, a needle body 1302 is provided in view (a), and a recess 1306 is generated in a proximal end of the needle body 1302 in view (b).

With specific reference to the prior art (non-RFID-equipped) suture needle 1300 of FIGS. 19a-d, a suture thread 1312 is inserted into the recess 1306 in view (c) and the suture thread 1312 is crimped (represented by arrows "C") in view (d). The prior art suture needle 1300 is then complete.

FIGS. 20a-22f schematically depict assembly of the RFID-enabled surgical suture needles of FIGS. 13, 14, and 15, respectively. In each of FIGS. 20a-22f, view (d) depicts the insertion of the RFID chip 1308 and inductive loop 1310 into the recess 1306, a suture thread 1312 is inserted into the recess 1306 in view (e) and the suture thread 1312 is crimped (represented by arrows "C") in view (f) to complete the suture needle 1300 apparatus. The assembly differences in FIGS. 20a-22f occur in view (c).

In FIG. 20c, the nonconductive shell 1314 is provided to the recess 1306 to receive the RFID chip 1308 and the inductive loop 1310. In FIG. 21c, the nonconductive skin 1416 is provided to at least the recess 1306' and optionally to the entire needle body 1302'. In FIG. 22c, the plug-type nonconductive shell 1314" is provided to the recess 1306 to receive the RFID chip 1308 and the inductive loop 1310.

In FIGS. 23a-h, a needle body 1302 is provided in view (a), and a recess 1306 is generated in a proximal end of the needle body 1302 in view (b). However, due to the aforementioned differences between the aspects of FIGS. 13-16 and FIGS. 17-18, FIG. 23c depicts the suture thread 1312" being inserted into the recess 1306", and crimped into place (arrows "C") in views (d)-(e). Then, in FIG. 23f, a washer 1720 is slid down over the suture thread 1312" and optionally into contact with a rim 1722 of the recess 1306". FIG. 23g shows the annular RFID chip 1308" (and any associated inductive loops 1310") being slid down over the suture thread 1312" and into contact with any washer 1720 that was provided. Finally, in FIG. 2h3, the cap 1724 is installed to cover the washer 1720 (when present) and annular RFID chip 1308" to complete the assembly.

It is contemplated that the antenna, whether associated with the needle body 1302 or the conductive suture thread 1312 will be electrically isolated from the inductive loop on the RFID chip 1308, for most use environments. Complete electrical isolation between the needle 1300 and conductive thread 1312 may be desired in some use environments, as well, but may be optional in others.

While the RFID chip 1308 is disclosed as being disposed axially between the needle body 1302 and the suture thread 1312 (i.e., at the swage end of the needle), it is contemplated that any RFID chip(s), of any examples or aspects of the technology described herein, could be located as desired anywhere in/on the needle and/or thread including, but not limited to, at the mid-section or pointed tip of the needle.

Various aspects of the present technology are described below, in claim-mimicking format:

Aspect 1. A suture needle apparatus comprising: a needle body;

a radio-frequency identification ("RFID") chip disposed proximate to the needle body, the RFID chip encoded with an identifying information associated with the suture needle apparatus;

an antenna spanning at least a portion of the needle body and configured to wirelessly communicate the identifying information; and

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a sheath comprising radio frequency ("RF") lucent material coextensive with and at least partially enclosing the RFID chip and the antenna.

Aspect 2. The suture needle apparatus of aspect 1, wherein the sheath encloses at least a portion of the needle body.

Aspect 3. The suture needle apparatus of aspect 1, wherein the antenna comprises at least a portion of the needle body.

Aspect 4. The suture needle apparatus of aspect 1, wherein the RFID chip is disposed inside the needle body.

Aspect 5. The suture needle apparatus of aspect 1, wherein the RFID chip is potted inside a recess of the needle body.

Aspect 6. The suture needle apparatus of aspect 1, wherein the needle body is at least one of partially metallic, organic, plastic, and composite.

Aspect 7. The suture needle apparatus of aspect 1, wherein the antenna at least partially envelops the body and RFID chip.

Aspect 8. The suture needle apparatus of aspect 1, further comprising a suture extending from the needle body.

Aspect 9. The suture needle apparatus of aspect 8, further comprising an eye loop for receiving the suture.

Aspect 10. The suture needle apparatus of aspect 8, further comprising a swaged end coupled to the suture.

Aspect 11. The suture needle apparatus of aspect 8, wherein at least a portion of the suture comprises conductive material to provide a conductive suture, wherein the conductive suture is electrically isolated from the needle body, and wherein the conductive suture and needle body are connected to opposite terminals of a chip, thereby forming separate arms of the antenna.

Aspect 12. The suture needle apparatus of aspect 8, wherein the RFID chip spans the needle body and the suture.

Aspect 13. The suture needle apparatus of aspect 12, wherein the RFID chip is disposed axially between the needle body and the suture.

Aspect 14. The suture needle apparatus of aspect 13, wherein the RFID chip is wrapped around the suture.

Aspect 15. The suture needle apparatus of aspect 14, including: an outer layer wrapped around the RFID chip between the RFID chip and the needle body to connect the needle body to the RFID chip; and

an inner layer wrapped around the suture between the suture and the RFID chip to connect the suture to the RFID chip.

Aspect 16. The needle apparatus of aspect 1, wherein identifying information comprises at least one of suture size, type, lot, unique identifier, and expiration date.

Aspect 17. A suture needle apparatus comprising: a metallic needle body;

a suture extending from the metallic body, wherein the suture comprises conductive material,

a radio-frequency identification ("RFID") chip disposed axially between the body and the suture, the RFID chip encoded with an identifying information associated with the suture needle apparatus;

an antenna, comprising at least a portion of the metallic body, linked to the RFID chip, the antenna configured to wirelessly communicate the identifying information; and a protective sheath comprising RF lucent material, enclosing the RFID chip and antenna;

wherein the conductive suture is electrically isolated from the sheath, and wherein the conductive suture is coupled to the antenna, thereby forming an extension of the antenna.

Aspect 18. The suture needle apparatus of aspect 17, including:

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an outer layer disposed between the RFID chip and the needle body to connect the needle body to the RFID chip; and

an inner layer disposed between the suture and the RFID chip to connect the suture to the RFID chip.

Aspect 19. The suture needle of aspect 17, wherein the suture is electrically isolated from the protective sheath, and wherein the conductive suture is coupled to the antenna, thereby forming an extension of the antenna.

Aspect 20. A method for manufacturing a suture needle apparatus, the method comprising: providing a needle body; encoding a radio-frequency identification ("RFID") chip with information associated with the suture needle apparatus;

disposing the RFID chip proximate to the needle body; linking an antenna to the RFID chip, the antenna configured to wirelessly communicate the identifying information; and at least partially enclosing the RFID chip and the antenna with a sheath comprising RF lucent material.

Aspect 21. The method of aspect 20, wherein disposing the RFID chip proximate to the needle body includes disposing the RFID chip inside a recess of the needle body.

Aspect 22. The method of aspect 20, including swaging the suture needle with a suture.

Aspect 23. The method of aspect 22, including at least partially coating the suture with a conductive material.

Aspect 24. The method of aspect 23, including circumferentially wrapping the RFID chip around the suture.

Unless defined otherwise, all technical and scientific terms used herein have the same meaning as is commonly understood by one of skill in the art to which the present disclosure pertains.

To the extent that the term "includes" or "including" is used in the specification or the claims, it is intended to be inclusive in a manner similar to the term "comprising" as that term is interpreted when employed as a transitional word in a claim. Furthermore, to the extent that the term "or" is employed (e.g., A or B) it is intended to mean "A or B or both." When the applicants intend to indicate "only A or B but not both" then the term "only A or B but not both" will be employed. Thus, use of the term "or" herein is the inclusive, and not the exclusive use. Also, to the extent that the terms "in" or "into" are used in the specification or the claims, it is intended to additionally mean "on" or "onto." Furthermore, to the extent the term "connect" is used in the specification or claims, it is intended to mean not only "directly connected to," but also "indirectly connected to" such as connected through another component or components.

As used herein, the singular forms "a," "an" and "the" can include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms "comprises" and/or "comprising," as used herein, can specify the presence of stated features, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, steps, operations, elements, components, and/or groups thereof.

As used herein, the phrase "at least one of X and Y" can be interpreted to include X, Y, or a combination of X and Y. For example, if an element is described as having at least one of X and Y, the element may, at a particular time, include X, Y, or a combination of X and Y, the selection of which could vary from time to time. In contrast, the phrase "at least one of X" can be interpreted to include one or more Xs. It will be understood that when an element is referred to as being "on," "attached" to, "connected" to, "coupled" with, "contacting," etc., another element, it can be directly on, attached

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to, connected to, coupled with or contacting the other element or intervening elements may also be present. In contrast, when an element is referred to as being, for example, "directly on," "directly attached" to, "directly connected" to, "directly coupled" with or "directly contacting" another element, there are no intervening elements present. It will also be appreciated by those of skill in the art that references to a structure or feature that is disposed "directly adjacent" another feature may have portions that overlap or underlie the adjacent feature, whereas a structure or feature that is disposed "adjacent" another feature might not have portions that overlap or underlie the adjacent feature.

Spatially relative terms, such as "under," "below," "lower," "over," "upper" and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms can encompass different orientations of a device in use or operation, in addition to the orientation depicted in the figures. For example, if a device in the figures is inverted, elements described as "under" or "beneath" other elements or features would then be oriented "over" the other elements or features.

It will be understood that, although the terms "first," "second," etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. Thus, a "first" element discussed below could also be termed a "second" element without departing from the teachings of the present disclosure. The sequence of operations (or steps) is not limited to the order presented in the claims or figures unless specifically indicated otherwise.

While aspects of this disclosure have been particularly shown and described with reference to the example aspects above, it will be understood by those of ordinary skill in the art that various additional aspects may be contemplated. For example, the specific methods described above for using the apparatus are merely illustrative; one of ordinary skill in the art could readily determine any number of tools, sequences of steps, or other means/options for placing the above-described apparatus, or components thereof, into positions substantively similar to those shown and described herein. In an effort to maintain clarity in the Figures, certain ones of duplicative components shown have not been specifically numbered, but one of ordinary skill in the art will realize, based upon the components that were numbered, the element numbers which should be associated with the unnumbered components; no differentiation between similar components is intended or implied solely by the presence or absence of an element number in the Figures. Any of the described structures and components could be integrally formed as a single unitary or monolithic piece or made up of separate sub-components, with either of these formations involving any suitable stock or bespoke components and/or any suitable material or combinations of materials; however, the chosen material(s) should be biocompatible for many applications. Any of the described structures and components could be disposable or reusable as desired for a particular use environment. Any component could be provided with a user-perceptible marking to indicate a material, configuration, at least one dimension, or the like pertaining to that component, the user-perceptible marking potentially aiding a user in selecting one component from an array of similar components for a particular use environment. A "predetermined" status may be determined at any time before the structures being manipulated actually reach that

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status, the “predetermination” being made as late as immediately before the structure achieves the predetermined status. The term “substantially” is used herein to indicate a quality that is largely, but not necessarily wholly, that which is specified—a “substantial” quality admits of the potential for some relatively minor inclusion of a non-quality item. Though certain components described herein are shown as having specific geometric shapes, all structures of this disclosure may have any suitable shapes, sizes, configurations, relative relationships, cross-sectional areas, or any other physical characteristics as desirable for a particular application. It should be understood that elements shown as a single component may be replaced with multiple components, and elements shown as multiple components may be replaced with a single component. The drawings are not to scale and the proportion of certain elements may be exaggerated for the purpose of illustration. Any structures or features described with reference to one aspect or configuration could be provided, singly or in combination with other structures or features, to any other aspect or configuration, as it would be impractical to describe each of the aspects and configurations discussed herein as having all of the options discussed with respect to all of the other aspects and configurations. A device or method incorporating any of these features should be understood to fall under the scope of this disclosure as determined based upon the claims below and any equivalents thereof.

Other aspects, objects, and advantages can be obtained from a study of the drawings, the disclosure, and the appended claims.

We claim:

1. A suture needle apparatus comprising:
a needle body;
a radio-frequency identification (“RFID”) chip disposed proximate to the needle body, the RFID chip encoded with an identifying information associated with the suture needle apparatus, and the RFID chip including an electromagnetic coupling element; and
a suture thread operatively coupled to the needle body; wherein the RFID chip is disposed axially between the needle body and the suture thread;
wherein the RFID chip and at least a portion of a suture thread are concurrently received within a recess of the needle body, the RFID chip being longitudinally interposed between the suture thread and at least a portion of the needle body; and
wherein at least one of the needle body and the suture thread is an antenna selectively electromagnetically coupled to the RFID chip and, when coupled, is configured to wirelessly communicate the identifying information responsive to radio-frequency interrogation of the suture needle apparatus.
2. The suture needle apparatus of claim 1, wherein the RFID chip is at least partially disposed inside a recess of the needle body.
3. The suture needle apparatus of claim 2, wherein the needle body includes a recess configured to concurrently receive the RFID chip and at least a portion of the suture thread.
4. The suture needle apparatus of claim 1, wherein the needle body is at least one of partially metallic, organic, plastic, and composite.
5. The suture needle apparatus of claim 1, wherein the suture thread extends from the needle body.
6. The suture needle apparatus of claim 5, wherein the needle body includes a swaged end coupled to the suture thread.

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7. The suture needle apparatus of claim 6, wherein the swaged end is crimped to directly connect the needle body and the suture thread.

8. The suture needle apparatus of claim 1, wherein at least a portion of the suture comprises conductive material to provide a conductive suture thread, wherein the conductive suture thread is electrically isolated from the needle body, and wherein the conductive suture thread is at least a portion of the antenna.

9. The needle apparatus of claim 1, wherein identifying information comprises at least one of suture size, type, lot, unique identifier, and expiration date.

10. The suture needle apparatus of claim 1, wherein the RFID chip is at least partially enclosed within a nonconductive shell configured to electrically isolate the RFID chip from the needle body.

11. The suture needle apparatus of claim 10, wherein the needle body includes a recess for selectively receiving the RFID chip, and wherein the nonconductive shell closely conforms to at least one inner dimension of the recess.

12. The suture needle apparatus of claim 11, wherein the nonconductive shell is selectively frictionally engaged into the recess.

13. The suture needle apparatus of claim 1, wherein the needle body is at least partially coated with a nonconductive skin configured to electrically isolate the RFID chip from the needle body.

14. The suture needle apparatus of claim 1, wherein the RFID chip has an annular profile including a center aperture configured to selectively receive at least a portion of the suture thread.

15. The suture needle apparatus of claim 14, wherein the needle body includes a recess for selectively concurrently receiving the RFID chip and a portion of the suture thread, the needle apparatus including a cap fitting about a circumference of the suture thread for preventing egress of the RFID chip from the suture needle.

16. The suture needle apparatus of claim 1, wherein the electromagnetic coupling is inductive coupling.

17. A method of providing a suture needle apparatus, the method comprising:

- providing a needle body;
- providing a radio-frequency identification (“RFID”) chip including an electromagnetic coupling element;
- encoding the RFID chip with information associated with the suture needle apparatus;
- disposing the RFID chip proximate to the needle body;
- operatively coupling a suture thread to the needle body with the RFID chip disposed axially between the needle body and the suture thread, including concurrently receiving the RFID chip and at least a portion of a suture thread within a recess of the needle body, the RFID chip being longitudinally interposed between the suture thread and at least a portion of the needle body;
- selectively electromagnetically coupling at least one of the needle body and the suture thread to the RFID chip as an antenna; and
- when the antenna and the RFID chip are electromagnetically coupled, wirelessly communicating the identifying information responsive to radio-frequency interrogation of the suture needle apparatus.

18. The method of claim 17, including electrically isolating the RFID chip from the needle body.

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19. The method of claim **18**, including
providing the RFID chip with an annular profile including
a center aperture;
extending at least a portion of the suture thread through
the center aperture; 5
concurrently with the RFID chip surrounding at least a
portion of the suture thread, placing an other portion of
the suture thread into a recess of the needle body; and
with a cap fit about a circumference of the suture thread,
preventing egress of the RFID chip from the suture 10
needle.

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